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(54) **AUTHENTICATED READING OF MEMORY SYSTEM DATA**

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G06F 12/14 (2006.01)
G06F 21/79 (2013.01)

(52) **U.S. Cl.**
CPC **G06F 12/1408** (2013.01); **G06F 12/145** (2013.01); **G06F 12/1466** (2013.01); **G06F 21/79** (2013.01)

(58) **Field of Classification Search**
CPC G06F 12/1408; G06F 12/145; G06F 12/1466; G06F 21/79
See application file for complete search history.

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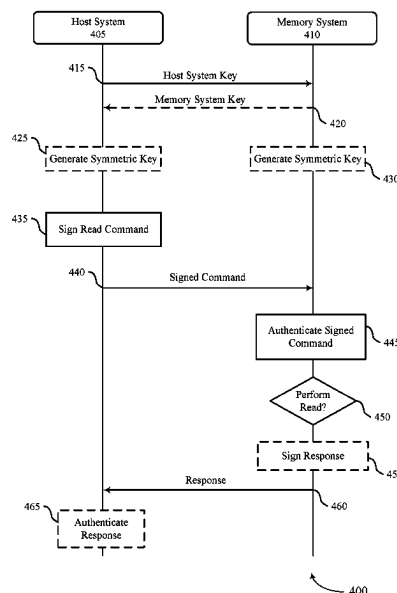
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(57) **ABSTRACT**

Methods, systems, and devices for authenticated reading of memory system data are described. In some examples, a host system and a memory system may exchange keys used to grant the host system access to one or more protected regions of the memory system. The keys may be symmetric or asymmetric. In some cases, the host system may transmit a read command to access data stored at a protected region of the memory system, along with a signature generated using the key associated with the protected region. The memory system may verify the signature to determine whether the host is authorized to access the protected region, and may transmit the requested data to the host system. In some examples, the memory system may sign the returned data, so that the host system may verify the source of the data.

23 Claims, 8 Drawing Sheets



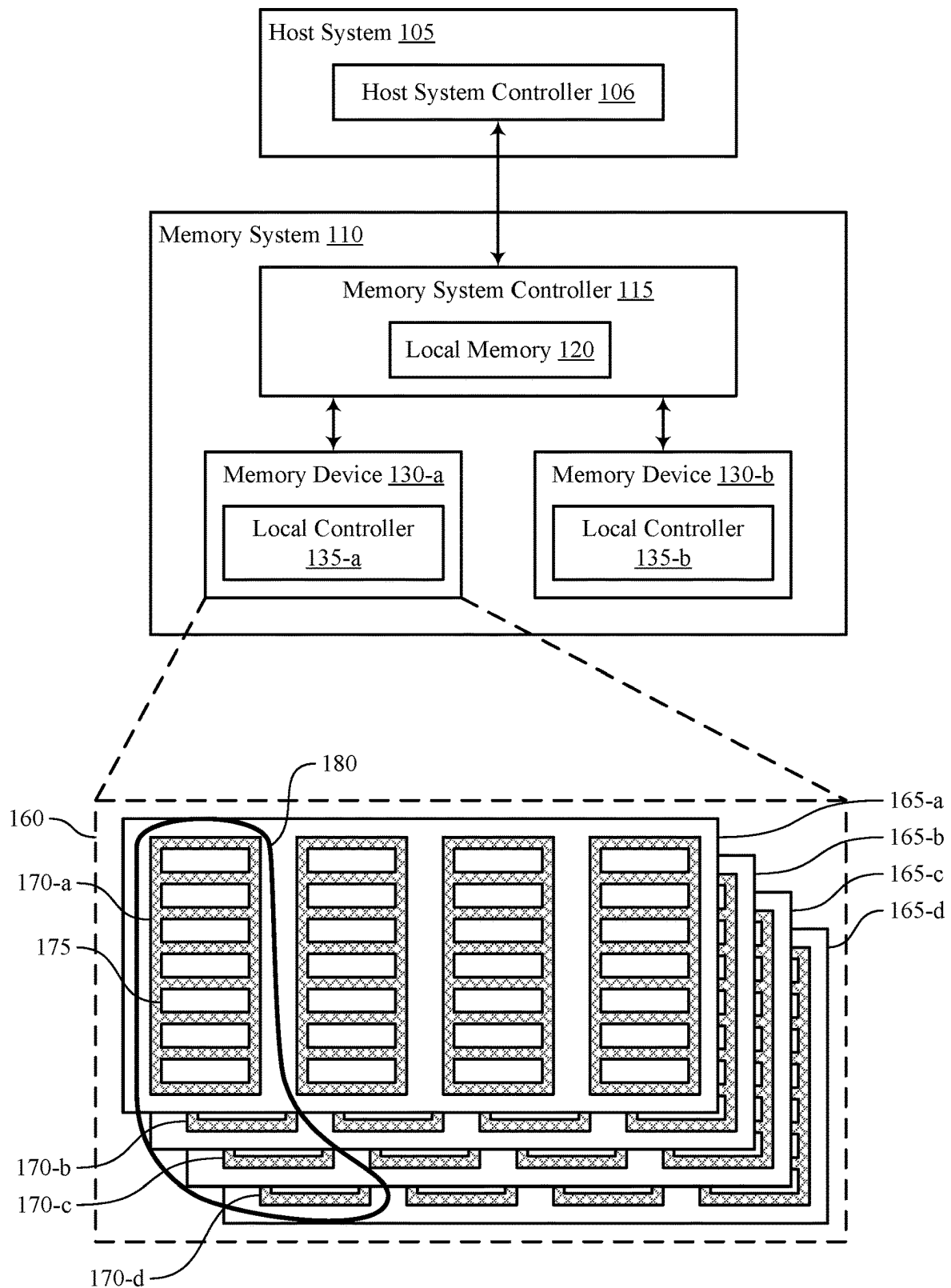


FIG. 1

100

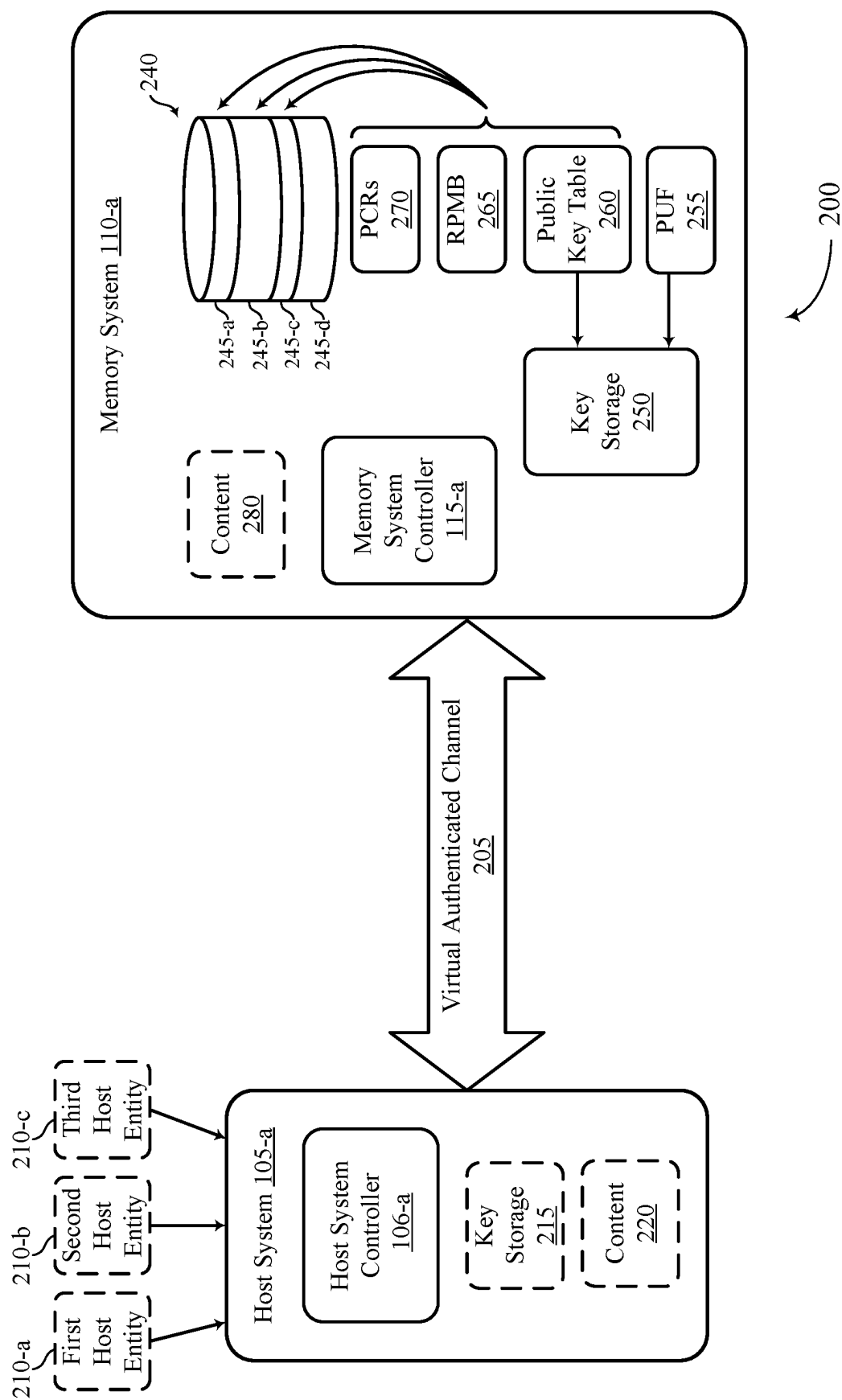
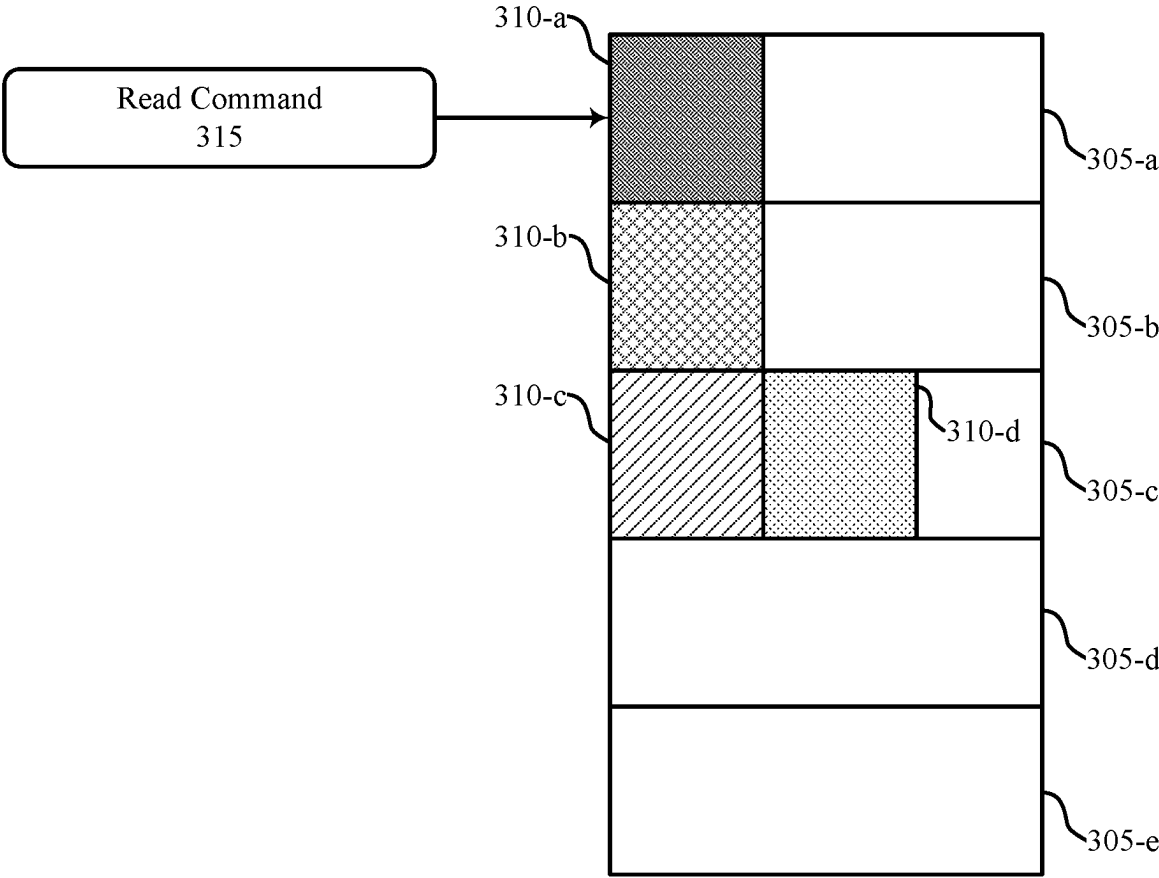


FIG. 2



300

FIG. 3

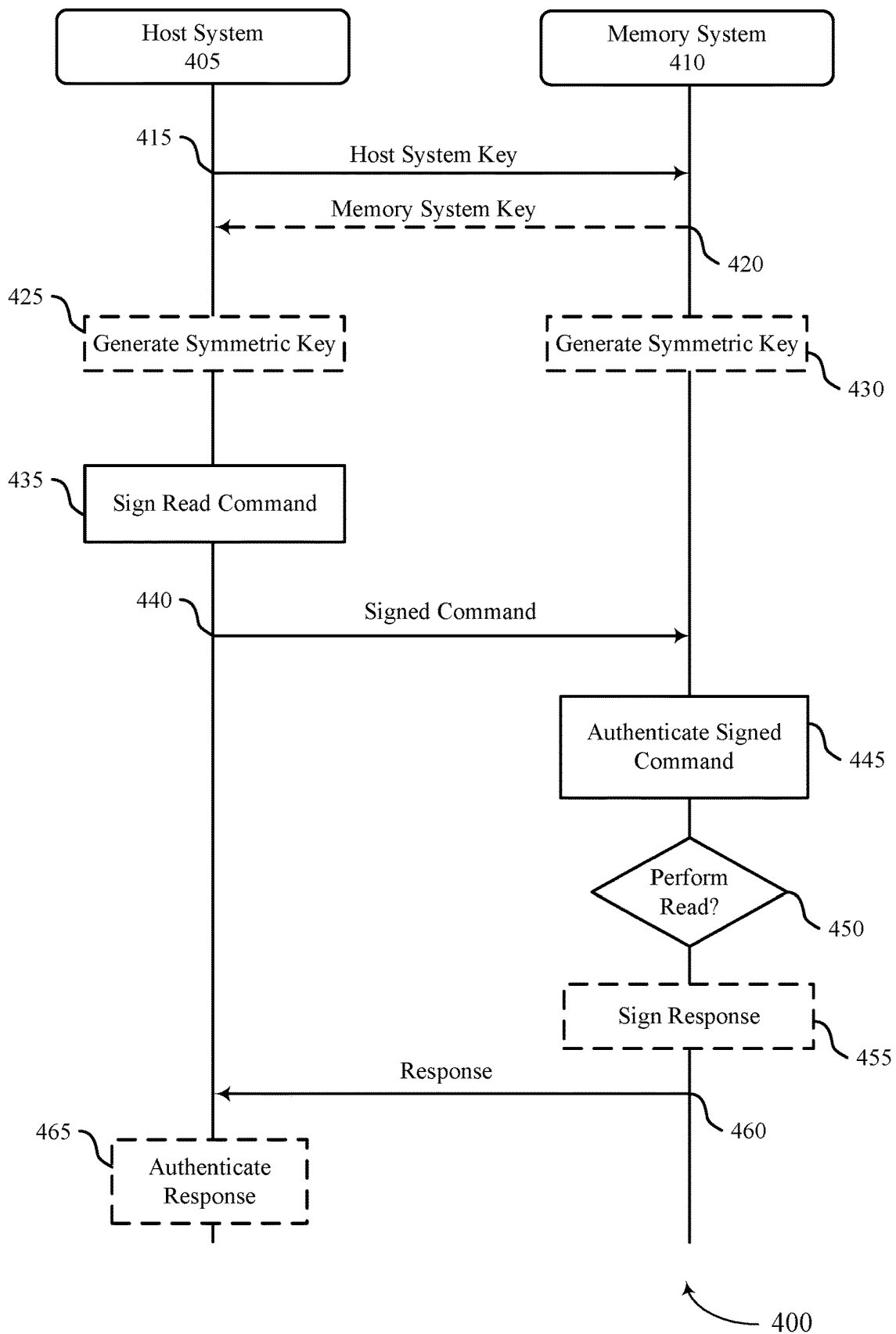


FIG. 4

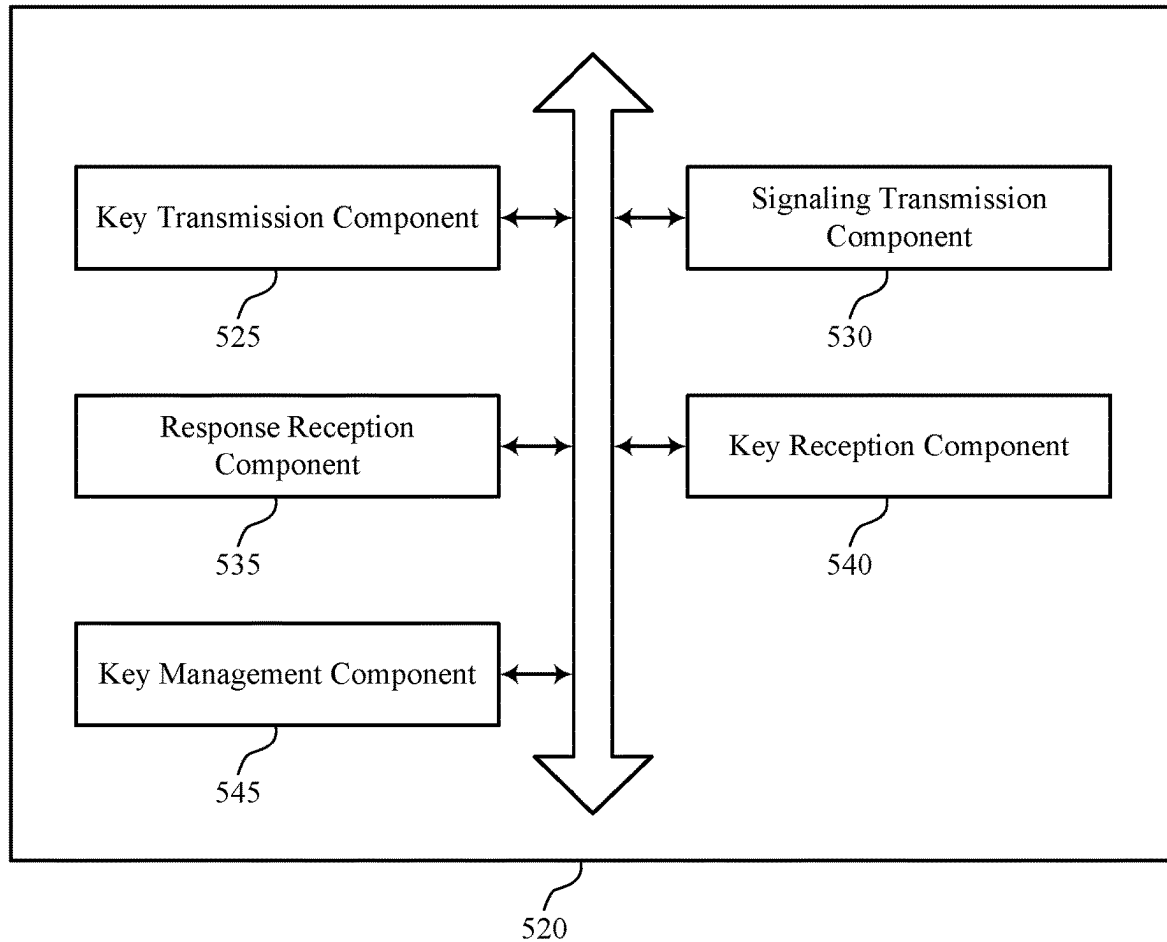
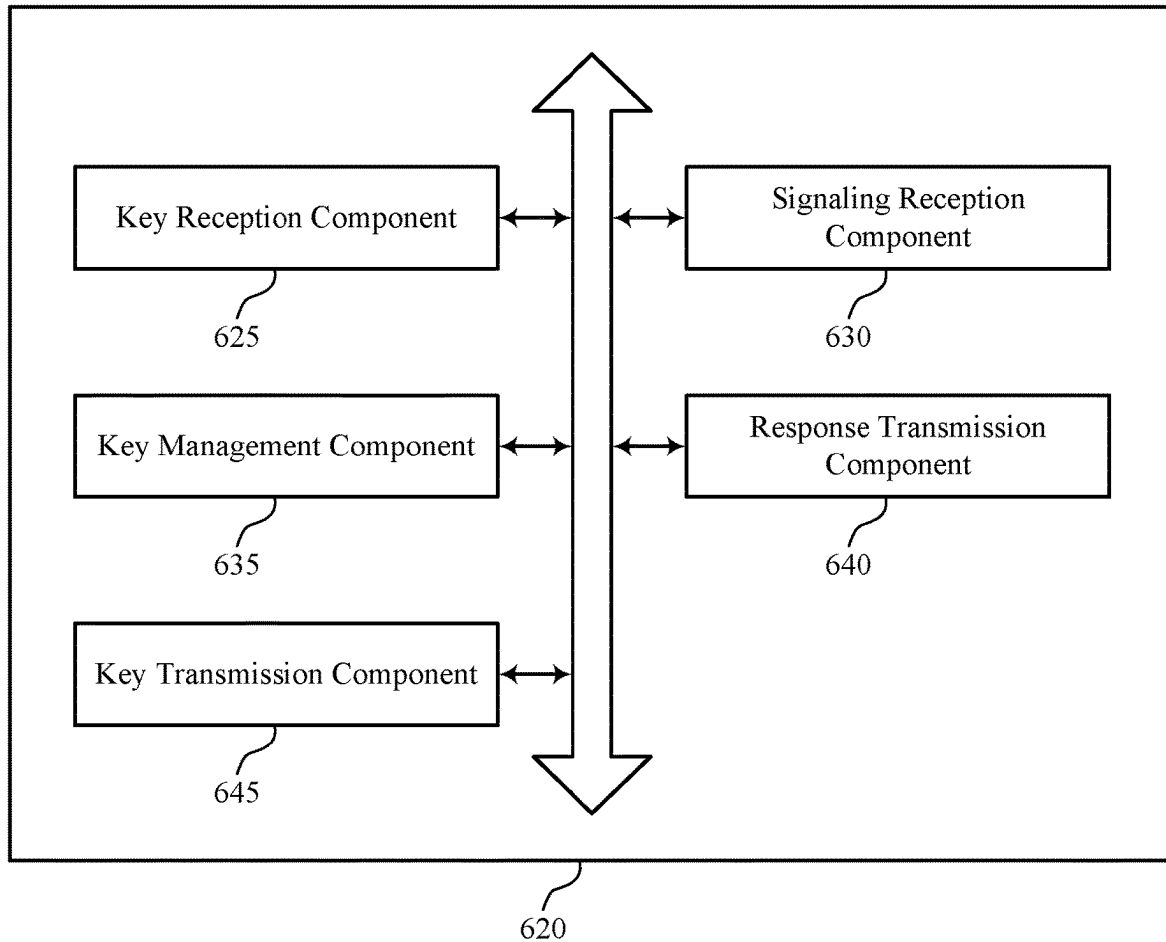
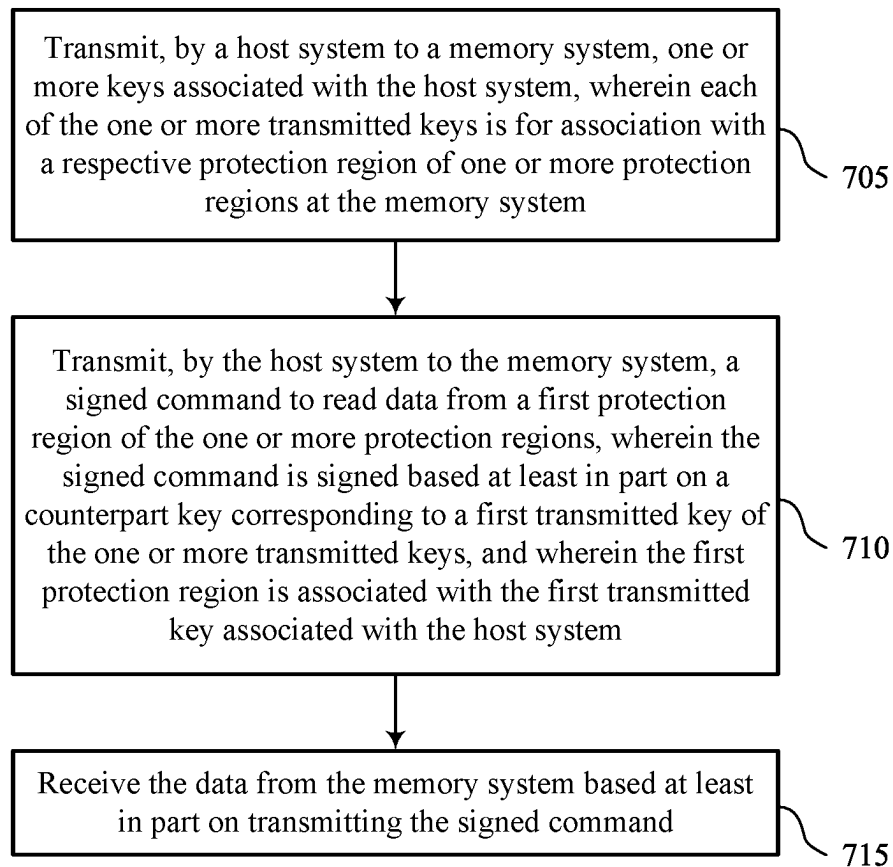


FIG. 5



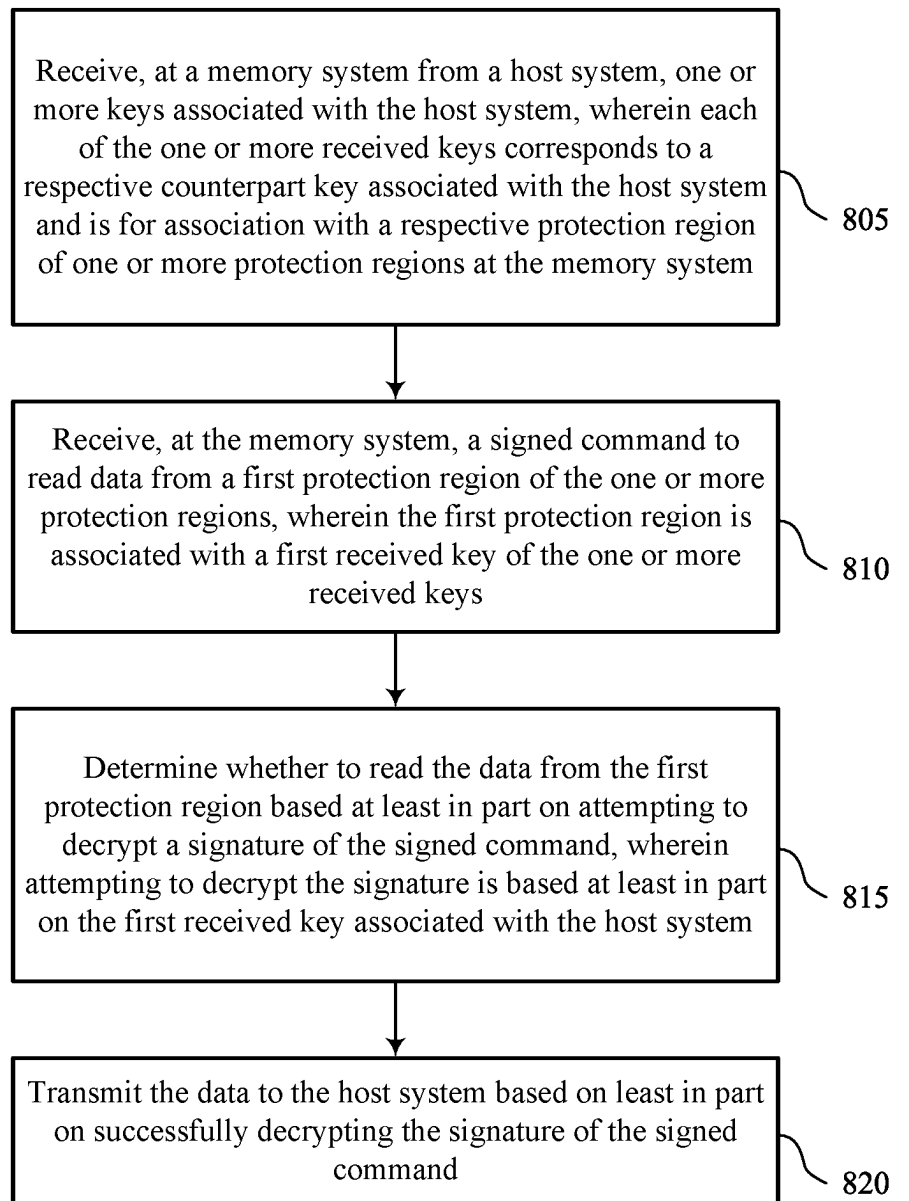
600

FIG. 6



700

FIG. 7



800

FIG. 8

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AUTHENTICATED READING OF MEMORY SYSTEM DATA

CROSS REFERENCE

The present application for patent claims the benefit of U.S. Provisional Patent Application No. 63/270,927 by DOVER, entitled "AUTHENTICATED READING OF MEMORY SYSTEM DATA," filed Oct. 22, 2021, assigned to the assignee hereof, and expressly incorporated by reference herein.

FIELD OF TECHNOLOGY

The following relates generally to one or more systems for memory and more specifically to authenticated reading of memory system data.

BACKGROUND

Memory devices are widely used to store information in various electronic devices such as computers, user devices, wireless communication devices, cameras, digital displays, and the like. Information is stored by programming memory cells within a memory device to various states. For example, binary memory cells may be programmed to one of two supported states, often corresponding to a logic 1 or a logic 0. In some examples, a single memory cell may support more than two possible states, any one of which may be stored by the memory cell. To access information stored by a memory device, a component may read, or sense, the state of one or more memory cells within the memory device. To store information, a component may write, or program, one or more memory cells within the memory device to corresponding states.

Various types of memory devices exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), static RAM (SRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), 3-dimensional cross-point memory (3D cross point), not-or (NOR) and not-and (NAND) memory devices, and others. Memory devices may be volatile or non-volatile. Volatile memory cells (e.g., DRAM cells) may lose their programmed states over time unless they are periodically refreshed by an external power source. Non-volatile memory cells (e.g., NAND memory cells) may maintain their programmed states for extended periods of time even in the absence of an external power source.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example of a system that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

FIG. 2 illustrates an example of a system that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

FIG. 3 illustrates an example of a system that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

FIG. 4 illustrates an example of a process flow that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

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FIG. 5 shows a block diagram of a host device that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

FIG. 6 shows a block diagram of a memory device that supports authenticated reading of memory system data in accordance with examples as disclosed herein.

FIGS. 7 and 8 show flowcharts illustrating a method or methods that support authenticated reading of memory system data in accordance with examples as disclosed herein.

DETAILED DESCRIPTION

Some computing platforms may involve one or more host systems that are operable to communicate with (e.g., directly or indirectly) one or more memory systems. For example, a host system may transmit access commands, such as a read command, to a memory system to access data stored in a secure location, such as a replay protected memory block (RPMB) using pairs of symmetric keys stored at the host system and the memory system. In some cases, the memory system may sign the data associated with the access command to verify to the host system that the data originated from the memory. However, some approaches to performing read commands from protected regions of the memory device may include removing the protected status of the region while retrieving the data, which may leave the memory system vulnerable to malicious parties, which may attempt to access the data while the region is unprotected. Further, some approaches to protecting regions of the memory device, such as by using an RPMB, may not allow for more secure data protection schemes, such as using asymmetric keys to sign and verify data or updating host system or memory system keys over time. Additionally, some approaches may not allow custom configuration or dynamic adjustment of protected region attributes, such as size of the protected region or a range of addresses associated with the protected region. Efficient techniques to access protected regions of a memory device are desired.

As described in the present disclosure, a host system and a memory system may exchange keys used to grant the host system access to one or more protected regions of the memory system. The keys may be symmetric (e.g., the host system and the memory system may share a same key) or asymmetric (e.g., both the host system and the memory system may have a unique private key, and may each share a corresponding public key with the other), and may be updated periodically (e.g., according to a cryptoperiod determined by the host system or the memory system). Additionally, the host system and the memory system may exchange separate keys for different protected regions of the memory system.

In some cases, the host system may transmit a read command to access data stored at a protected region of the memory system, and the host system may sign the read command using the key associated with the protected region. Upon receiving the read command, the memory system may verify the signature to determine whether the host is authorized to access the protected region, and may transmit the requested data to the host system. In some examples, the memory system may sign the returned data, so that the host system may verify the source of the data. In some cases, the protected regions of the memory system may be updated, for example by adjusting the size or address range of the protected regions, in response to a command from the host system. Techniques described herein may increase the security of the computing platform, for example by more thoroughly protecting data from malicious parties

and allowing the source of data transmitted between the host system and the memory system to be verified.

Features of the disclosure are initially described in the context of systems, devices, and circuits with reference to FIG. 1. Features of the disclosure are described in the context of systems and a process flow with reference to FIGS. 2-4. These and other features of the disclosure are further illustrated by and described in the context of an apparatus diagram and flowchart that relate to authenticated reading of memory system data with reference to FIGS. 5-8.

FIG. 1 illustrates an example of a system 100 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The system 100 includes a host system 105 coupled with a memory system 110.

A memory system 110 may be or include any device or collection of devices, where the device or collection of devices includes at least one memory array. For example, a memory system 110 may be or include a Universal Flash Storage (UFS) device, an embedded Multi-Media Controller (eMMC) device, a flash device, a universal serial bus (USB) flash device, a secure digital (SD) card, a solid-state drive (SSD), a hard disk drive (HDD), a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), or a non-volatile DIMM (NVDIMM), among other possibilities.

The system 100 may be included in a computing device such as a desktop computer, a laptop computer, a network server, a mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), an Internet of Things (IoT) enabled device, an embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or any other computing device that includes memory and a processing device.

The system 100 may include a host system 105, which may be coupled with the memory system 110. In some examples, this coupling may include an interface with a host system controller 106, which may be an example of a controller or control component configured to cause the host system 105 to perform various operations in accordance with examples as described herein. The host system 105 may include one or more devices, and in some cases may include a processor chipset and a software stack executed by the processor chipset. For example, the host system 105 may include an application configured for communicating with the memory system 110 or a device therein. The processor chipset may include one or more cores, one or more caches (e.g., memory local to or included in the host system 105), a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., peripheral component interconnect express (PCIe) controller, serial advanced technology attachment (SATA) controller). The host system 105 may use the memory system 110, for example, to write data to the memory system 110 and read data from the memory system 110. Although one memory system 110 is shown in FIG. 1, the host system 105 may be coupled with any quantity of memory systems 110.

The host system 105 may be coupled with the memory system 110 via at least one physical host interface. The host system 105 and the memory system 110 may in some cases be configured to communicate via a physical host interface using an associated protocol (e.g., to exchange or otherwise communicate control, address, data, and other signals between the memory system 110 and the host system 105). Examples of a physical host interface may include, but are not limited to, a SATA interface, a UFS interface, an eMMC interface, a PCIe interface, a USB interface, a Fiber Channel interface, a Small Computer System Interface (SCSI), a

Serial Attached SCSI (SAS), a Double Data Rate (DDR) interface, a DIMM interface (e.g., DIMM socket interface that supports DDR), an Open NAND Flash Interface (ONFI), and a Low Power Double Data Rate (LPDDR) interface. In some examples, one or more such interfaces may be included in or otherwise supported between a host system controller 106 of the host system 105 and a memory system controller 115 of the memory system 110. In some examples, the host system 105 may be coupled with the memory system 110 (e.g., the host system controller 106 may be coupled with the memory system controller 115) via a respective physical host interface for each memory device 130 included in the memory system 110, or via a respective physical host interface for each type of memory device 130 included in the memory system 110.

The memory system 110 may include a memory system controller 115 and one or more memory devices 130. A memory device 130 may include one or more memory arrays of any type of memory cells (e.g., non-volatile memory cells, volatile memory cells, or any combination thereof). Although two memory devices 130-a and 130-b are shown in the example of FIG. 1, the memory system 110 may include any quantity of memory devices 130. Further, if the memory system 110 includes more than one memory device 130, different memory devices 130 within the memory system 110 may include the same or different types of memory cells.

The memory system controller 115 may be coupled with and communicate with the host system 105 (e.g., via the physical host interface) and may be an example of a controller or control component configured to cause the memory system 110 to perform various operations in accordance with examples as described herein. The memory system controller 115 may also be coupled with and communicate with memory devices 130 to perform operations such as reading data, writing data, erasing data, or refreshing data at a memory device 130—among other such operations—which may generically be referred to as access operations. In some cases, the memory system controller 115 may receive commands from the host system 105 and communicate with one or more memory devices 130 to execute such commands (e.g., at memory arrays within the one or more memory devices 130). For example, the memory system controller 115 may receive commands or operations from the host system 105 and may convert the commands or operations into instructions or appropriate commands to achieve the desired access of the memory devices 130. In some cases, the memory system controller 115 may exchange data with the host system 105 and with one or more memory devices 130 (e.g., in response to or otherwise in association with commands from the host system 105). For example, the memory system controller 115 may convert responses (e.g., data packets or other signals) associated with the memory devices 130 into corresponding signals for the host system 105.

The memory system controller 115 may be configured for other operations associated with the memory devices 130. For example, the memory system controller 115 may execute or manage operations such as wear-leveling operations, garbage collection operations, error control operations such as error-detecting operations or error-correcting operations, encryption operations, caching operations, media management operations, background refresh, health monitoring, and address translations between logical addresses (e.g., logical block addresses (LBAs)) associated with commands from the host system 105 and physical addresses

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(e.g., physical block addresses) associated with memory cells within the memory devices **130**.

The memory system controller **115** may include hardware such as one or more integrated circuits or discrete components, a buffer memory, or a combination thereof. The hardware may include circuitry with dedicated (e.g., hard-coded) logic to perform the operations ascribed herein to the memory system controller **115**. The memory system controller **115** may be or include a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a digital signal processor (DSP)), or any other suitable processor or processing circuitry.

The memory system controller **115** may also include a local memory **120**. In some cases, the local memory **120** may include read-only memory (ROM) or other memory that may store operating code (e.g., executable instructions) executable by the memory system controller **115** to perform functions ascribed herein to the memory system controller **115**. In some cases, the local memory **120** may additionally or alternatively include static random access memory (SRAM) or other memory that may be used by the memory system controller **115** for internal storage or calculations, for example, related to the functions ascribed herein to the memory system controller **115**. Additionally or alternatively, the local memory **120** may serve as a cache for the memory system controller **115**. For example, data may be stored in the local memory **120** if read from or written to a memory device **130**, and the data may be available within the local memory **120** for subsequent retrieval for or manipulation (e.g., updating) by the host system **105** (e.g., with reduced latency relative to a memory device **130**) in accordance with a cache policy.

Although the example of the memory system **110** in FIG. **1** has been illustrated as including the memory system controller **115**, in some cases, a memory system **110** may not include a memory system controller **115**. For example, the memory system **110** may additionally or alternatively rely upon an external controller (e.g., implemented by the host system **105**) or one or more local controllers **135**, which may be internal to memory devices **130**, respectively, to perform the functions ascribed herein to the memory system controller **115**. In general, one or more functions ascribed herein to the memory system controller **115** may in some cases instead be performed by the host system **105**, a local controller **135**, or any combination thereof. In some cases, a memory device **130** that is managed at least in part by a memory system controller **115** may be referred to as a managed memory device. An example of a managed memory device is a managed NAND (MNAND) device.

A memory device **130** may include one or more arrays of non-volatile memory cells. For example, a memory device **130** may include NAND (e.g., NAND flash) memory, ROM, phase change memory (PCM), self-selecting memory, other chalcogenide-based memories, ferroelectric random access memory (RAM) (FeRAM), magnetoresistive RAM (MRAM), NOR (e.g., NOR flash) memory, Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), electrically erasable programmable ROM (EEPROM), or any combination thereof.

Additionally or alternatively, a memory device **130** may include one or more arrays of volatile memory cells. For example, a memory device **130** may include RAM memory cells, such as dynamic RAM (DRAM) memory cells and synchronous DRAM (SDRAM) memory cells.

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In some examples, a memory device **130** may include (e.g., on a same die or within a same package) a local controller **135**, which may execute operations on one or more memory cells of the respective memory device **130**. A local controller **135** may operate in conjunction with a memory system controller **115** or may perform one or more functions ascribed herein to the memory system controller **115**. For example, as illustrated in FIG. **1**, a memory device **130-a** may include a local controller **135-a** and a memory device **130-b** may include a local controller **135-b**.

In some cases, a memory device **130** may be or include a NAND device (e.g., NAND flash device). A memory device **130** may be or include a memory die **160**. For example, in some cases, a memory device **130** may be a package that includes one or more dies **160**. A die **160** may, in some examples, be a piece of electronics-grade semiconductor cut from a wafer (e.g., a silicon die cut from a silicon wafer). Each die **160** may include one or more planes **165**, and each plane **165** may include a respective set of blocks **170**, where each block **170** may include a respective set of pages **175**, and each page **175** may include a set of memory cells.

In some cases, a NAND memory device **130** may include memory cells configured to each store one bit of information, which may be referred to as single level cells (SLCs). Additionally or alternatively, a NAND memory device **130** may include memory cells configured to each store multiple bits of information, which may be referred to as multi-level cells (MLCs) if configured to each store two bits of information, as tri-level cells (TLCs) if configured to each store three bits of information, as quad-level cells (QLCs) if configured to each store four bits of information, or more generically as multiple-level memory cells. Multiple-level memory cells may provide greater density of storage relative to SLC memory cells but may, in some cases, involve narrower read or write margins or greater complexities for supporting circuitry.

In some cases, planes **165** may refer to groups of blocks **170**, and in some cases, concurrent operations may take place within different planes **165**. For example, concurrent operations may be performed on memory cells within different blocks **170** so long as the different blocks **170** are in different planes **165**. In some cases, an individual block **170** may be referred to as a physical block, and a virtual block **180** may refer to a group of blocks **170** within which concurrent operations may occur. For example, concurrent operations may be performed on blocks **170-a**, **170-b**, **170-c**, and **170-d** that are within planes **165-a**, **165-b**, **165-c**, and **165-d**, respectively, and blocks **170-a**, **170-b**, **170-c**, and **170-d** may be collectively referred to as a virtual block **180**.

In some cases, a virtual block may include blocks **170** from different memory devices **130** (e.g., including blocks in one or more planes of memory device **130-a** and memory device **130-b**). In some cases, the blocks **170** within a virtual block may have the same block address within their respective planes **165** (e.g., block **170-a** may be “block 0” of plane **165-a**, block **170-b** may be “block 0” of plane **165-b**, and so on). In some cases, performing concurrent operations in different planes **165** may be subject to one or more restrictions, such as concurrent operations being performed on memory cells within different pages **175** that have the same page address within their respective planes **165** (e.g., related to command decoding, page address decoding circuitry, or other circuitry being shared across planes **165**).

In some cases, a block **170** may include memory cells organized into rows (pages **175**) and columns (e.g., strings, not shown). For example, memory cells in a same page **175** may share (e.g., be coupled with) a common word line, and

memory cells in a same string may share (e.g., be coupled with) a common digit line (which may alternatively be referred to as a bit line).

For some NAND architectures, memory cells may be read and programmed (e.g., written) at a first level of granularity (e.g., at the page level of granularity) but may be erased at a second level of granularity (e.g., at the block level of granularity). That is, a page 175 may be the smallest unit of memory (e.g., set of memory cells) that may be independently programmed or read (e.g., programmed or read concurrently as part of a single program or read operation), and a block 170 may be the smallest unit of memory (e.g., set of memory cells) that may be independently erased (e.g., erased concurrently as part of a single erase operation). Further, in some cases, NAND memory cells may be erased before they can be re-written with new data. Thus, for example, a used page 175 may in some cases not be updated until the entire block 170 that includes the page 175 has been erased.

The system 100 may include any quantity of non-transitory computer readable media that support authenticated reading of memory system data. For example, the host system 105, the memory system controller 115, or a memory device 130 (e.g., a local controller 135) may include or otherwise may access one or more non-transitory computer readable media storing instructions (e.g., firmware) for performing the functions ascribed herein to the host system 105, memory system controller 115, or memory device 130. For example, such instructions, if executed by the host system 105 (e.g., by the host system controller 106), by the memory system controller 115, or by a memory device 130 (e.g., by a local controller 135), may cause the host system 105, memory system controller 115, or memory device 130 to perform one or more associated functions as described herein.

In some cases, a memory system 110 may utilize a memory system controller 115 to provide a managed memory system that may include, for example, one or more memory arrays and related circuitry combined with a local (e.g., on-die or in-package) controller (e.g., local controller 135). An example of a managed memory system is a managed NAND (MNAND) system.

In some examples, a host system 105 and a memory system 110 may exchange keys used to grant the host system 105 access to one or more protected regions of the memory system 110 (e.g., one or more protected regions of a memory device 130). The keys may symmetric (e.g., the host system 105 and the memory system 110 may share a same key) or asymmetric (e.g., both the host system 105 and the memory system 110 may have a unique private key, and may each share a corresponding public key with the other), and may be updated periodically (e.g., according to a cryptoperiod determined by the host system 105 or the memory system 110). Additionally, the host system 105 and the memory system 110 may exchange separate keys for different protected regions of the memory system 110. In some cases, the host system 105 may transmit a read command to access data stored at a protected region of the memory system 110, and the host system 105 may sign the read command using the key associated with the protected region. Upon receiving the read command, the memory system 105 may verify the signature to determine whether the host system 105 is authorized to access the protected region, and may transmit the requested data to the host system 105. In some examples, the memory system 110 may sign the returned data, so that the host system 105 may verify the source of the data. In some cases, the protected regions of the memory system 110

may be updated, for example by adjusting the size or address range of the protected regions, in response to a command from the host system. Techniques described herein may increase the security of the computing platform, for example by more thoroughly protecting data from malicious parties and allowing the source of data transmitted between the host system 105 and the memory system 110 to be verified.

FIG. 2 illustrates an example of a system 200 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The system 200 may include a host system 105-a and a memory system 110-a, which may be examples of the respective systems described with reference to FIG. 1. The host system 105-a and the memory system 110-a may implement various techniques for exchanging public keys to support the communication of signaling between the respective systems with identity authenticity (e.g., signatures) and integrity (e.g., encryption), among other characteristics, which may be based on unique and private cryptographic identities of the host system 105-a and the memory system 110-a. The host system 105-a may include a host system controller 106-a and the memory system 110-a may include a memory system controller 115-a, and, in some examples, the host system controller 106-a and the memory system controller 115-a may be configured to perform one or more of the described operations at the host system 105-a and the memory system 110-a, respectively. Although techniques are described with reference to a single host system 105-a and a single memory system 110-a of the system 200, the described techniques may be extended to support implementations of a host system 105 that is coupled with any quantity of memory systems 110, or implementations of a memory system 110 that is coupled with any quantity of host systems 105, or implementations of a network of multiple host systems 105 coupled with multiple memory systems 110.

The host system 105-a may be an example of a system that uses at least a portion of the memory system 110-a (e.g., storage 240) for information storage, which may include various operations that support the host system 105-a writing information to the memory system 110-a, or the host system 105-a reading information from the memory system 110-a, or both. In some examples, the host system 105-a may be characterized as being “local,” which may refer to a relatively direct or proximal physical, electrical, or otherwise communicative coupling. In some other examples, the host system 105-a may be characterized as being “remote,” which may refer to a relatively distant (e.g., non-co-located) communicative coupling that may involve one or more wired, wireless, optical, or otherwise relatively distant communicative couplings, such as a cloud application or otherwise distributed compute system.

In some examples, the host system 105-a may include, may be coupled with, or may be otherwise associated with one or more host entities 210. Host entities 210 may be implemented as hardware entities, firmware entities, or software entities, and may include various serial, parallel, or hierarchical coupling or logical organization with or via the host system 105-a. In some examples, the host entities 210 may request or otherwise perform signaling with the memory system 110-a via a common controller or interface (e.g., via host system controller 106-a). In various examples, host entities 210 may be associated with different functions, different feature sets, different permissions, different storage attributes (e.g., data protection attributes), among other different characteristics.

In some examples, each of the host entities 210 may be associated with a unique identifier (e.g., a secret identifier, a

unique device secret, a unique entity secret), which may include or may support the generation of a respective private key for the host entity **210**. In some examples, an identifier of a host entity **210** may not, itself, be private, but a private key may be generated (e.g., by the host system **105-a**) for a host entity **210** based on an identifier (e.g., public or private) of the host entity **210** and a private identifier of the host system **105-a** (e.g., a private master identifier). Such techniques may support each of the host entities **210** being uniquely identified and authenticated (e.g., separately from other host entities **210**) in accordance with examples as disclosed herein.

The example of host system **105-a** may be associated with an original equipment manufacturer (OEM) host entity **210-a** (e.g., a first host entity), an operating system (OS) vendor host entity **210-b** (e.g., a second host entity), and an independent software vendor (ISV) host entity **210-c** (e.g., a third host entity). In some other examples, a host system **105** may include or be otherwise associated with any quantity of one or more host entities **210** including but not limited to one or more OEM host entities **210**, OS vendor host entities **210**, ISV host entities **210**, or other types of host entities. In some examples, host entities **210** may be omitted or otherwise not separately considered, in which case a master private key may be implemented by the host system **105-a** (and any host entities **210**, where applicable), which may be based on a single or shared unique identifier of the host system **105-a** (e.g., a secret identifier, a unique device secret, or a unique host secret associated with the host system **105-a**).

In some examples, the host system **105-a** may be associated with a location for storing authentication or encryption information (e.g., generated or received keys, certificates), such as a key storage **215**. For example, the host system **105-a** may use the key storage **215** to store one or more private keys or certificates associated with the host system **105-a**. In some examples, a key storage **215** may be a portion of the host system **105-a**, such as an implementation of a dedicated storage component of the host system **105-a**. Additionally or alternatively, one or more components of the key storage **215** may be located outside the host system **105-a**, but may be otherwise accessible by the host system **105-a** (e.g., in a secure manner). In various examples, the key storage **215** may include a non-volatile storage location (e.g., for static keys or keys maintained for a relatively long time), or a volatile storage location (e.g., for ephemeral keys or keys that are otherwise generated relatively frequently), or both. Although the key storage **215** is illustrated separately from the host system controller **106-a**, in some examples, the key storage **215** may be part of or otherwise associated with the host system controller **106-a**, such as a storage location that also includes firmware for the host system **105-a** or the host system controller **106-a**.

In some examples, the host system **105-a** may include content **220**, which may refer to various types of information stored at the host system **105-a**. In some examples, content **220** may be accessed or otherwise used to support various key generation (e.g., content-based key generation) or other cryptographic techniques in accordance with examples as disclosed herein. For example, the content **220** may include firmware of the host system **105-a**, such as boot code (e.g., second-stage boot code, "L1" boot code), or a firmware security descriptor (FSD), which may be used to establish an operating or cryptographic state (e.g., a firmware state) of the host system **105-a**. In some examples, information associated with the content **220** may be transmitted to the memory system **110-a** to support various authentication or encryption techniques (e.g., for the memory system **110-a** to

generate keys or certificates for operation with the memory system **110-a**). Although the content **220** is illustrated separately from the host system controller **106-a**, in some examples, the content **220** may be part of or otherwise associated with the host system controller **106-a**, such as a storage location that includes firmware for the host system **105-a** or the host system controller **106-a**.

The memory system **110-a** may include storage **240**, which may refer to a collective storage capacity of one or more instances of local memory **120**, or of one or more memory devices **130**, or various combinations thereof that are included in or are otherwise associated with the memory system **110-a**. In some examples, the storage **240** may be divided or otherwise organized in partitions **245** (e.g., memory ranges, address ranges), which may refer to various subsets or ranges of logical addresses or physical addresses of the associated local memory **120** or memory devices **130**. In some examples, the partitions **245** may be assigned with an initial range of addresses, and may be updated with an assignment to a different range of addresses, including an appending of additional new addresses, an assignment to a subset of the initial range of addresses (e.g., a trimming of ranges), or an assignment to an entirely new range of addresses.

In some examples, partitions **245**, or portions thereof, may be assigned to or allocated to different functions or attributes, such as examples where one or more partitions **245** are associated with a respective one or more host entities **210**, or their respective public or private keys. In an example implementation, a partition **245-a** may be associated with the OEM host entity **210-c**, a partition **245-b** may be associated with the OS vendor host entity **210-b**, and a partition **245-c** may be associated with the ISV host entity **210-c**. In some examples, a partition **245-d** may be unallocated (e.g., not dedicated to a certain purpose or entity, free space), or may be shared among multiple host entities **210**, among other examples for allocating partitions **245**. In some examples, partitions **245** may be used to implement various hierarchical keying or authentication techniques. For example, each partition **245**, or some portion of a partition **245**, may be assigned with or updated with a protection attribute (e.g., enabling or disabling a write protection attribute, enabling or disabling a read protection attribute), which may be associated with various keys, authentications, or encryptions that are specific to a given host entity **210**, or that are common to the host system **105-a** in general, among other examples.

In some examples, the memory system **110-a** may be associated with a location for storing authentication or encryption information (e.g., generated or received keys, certificates), such as a key storage **250**. For example, the memory system **110-a** may use the key storage **250** to store one or more private keys associated with the memory system **110-a**, or one or more public keys or certificates generated by the memory system **110-a**, or one or more public keys or certificates received from the host system **105-a** (or other host systems **105**, not shown). In some examples, the key storage **250** may be a portion of the memory system **110-a**, such as an implementation of a dedicated storage component of the memory system **110-a**. Additionally or alternatively, the key storage **250** may be located outside the memory system **110-a**, but may be otherwise accessible by the memory system **110-a** (e.g., in a secure manner). In various examples, the key storage **250** may include a non-volatile storage location (e.g., for static keys or keys maintained for a relatively long time), or a volatile storage location (e.g., for ephemeral keys or keys that are otherwise generated rela-

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tively frequently), or both. Although the key storage **250** is illustrated separately from the storage **240**, the key storage **250** may, in some examples, be included in a portion of the storage **240** (e.g., in a separate or dedicated partition **245**). Further, although the key storage **250** is illustrated separately from the memory system controller **115-a**, in some examples, the key storage **250** may be part of or otherwise associated with the memory system controller **115-a**, such as a storage location that also includes firmware for the memory system **110-a** or the memory system controller **115-a** (e.g., a local memory **120**).

In some examples, the memory system **110-a** may include a physical unclonable function (PUF) **255**, which may support the assignment of or generation of an identifier that is unique to the memory system **110-a** (e.g., for generating a secret identifier or a unique device secret of the memory system **110-a**). The PUF **255** may include various components or circuit elements that have an intrinsic physical characteristic that is unique to the PUF **255**, which may be leveraged to establish an intrinsic uniqueness of the memory system **110-a**. For example, the PUF may include a set of one or more transistors, resistors, capacitors, memory cells (e.g., SRAM cells, which may, in some cases, be included in local memory **120** described with reference to FIG. 1), or other circuit elements or combination thereof which, when accessed, support the generation of a digital signature that is unique to the memory system **110-a**. In some examples, a controller of the memory system **110-a** (e.g., the memory system controller **115-a**) may access or otherwise interact with the PUF **255** to generate one or more private keys for the memory system **110-a**, which may subsequently be used to generate public keys for establishing authenticity or encryption between the memory system **110-a** and the host system **105-a** (e.g., or the host entities **210**, where applicable). Although the PUF **255** is illustrated as being separate from the key storage **250**, in some examples, the PUF **255** may be included in or be otherwise interpreted as being part of the key storage **250** (e.g., part of the memory system controller **115-a**, part of a local memory **120** of the memory system **110-a**).

In various implementations, the PUF **255** itself, or signaling generated by the PUF **255**, or both may be inaccessible from outside the memory system **110-a**. Such inaccessibility may be supported by various implementations of including the PUF **255**, and other components involved in the described cryptographic techniques, in a portion of the memory system **110-a** where attempts to access such components would be destructive to the components, or where such components or associated signaling are otherwise shielded from destructive or non-destructive probing or snooping techniques. For example, at least the PUF **255** and the other components involved in the described cryptographic techniques (e.g., components involved in handling private keys or unique device secrets, which may include at least a portion of the memory system controller **115-a** or at least some portion thereof), if not all the components of the memory system **110-a**, may be implemented in a contiguous semiconductor chip such as an SoC implementation.

In some examples, the memory system **110-a** may include a public key table **260** (e.g., an elliptical curve cryptography public key table), which may be configured to store, organize, or allocate public keys such as those received from the host system **105-a**, or those generated at the memory system **110-a**, or both. In some examples (e.g., in implementations where host entities **210** are associated with respective public keys that are transmitted by the host system **105-a**), the public key table **260** may hold a respective public key, or

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mapping thereof, for each of the OEM host entity **210-a**, the OS vendor host entity **210-b**, and the ISV host entity **210-c** (e.g., associated with the partitions **245-a**, **245-b**, and **245-c**, respectively). Although the public key table **260** is illustrated as being separate from the key storage **250**, in some examples, the public key table **260** may be included in or be otherwise interpreted as being part of the key storage **250** (e.g., part of the memory system controller **115-a**, part of a local memory **120** of the memory system **110-a**).

In some implementations, the public key table **260** may be associated with a mapping between public keys and device identifiers, or partitions **245**, or protection attributes (e.g., write protection configurations, read protection configurations), or various combinations thereof, among other mapping between keys and associated configurations. For example, the public key table **260** may provide a mapping for one or more host systems **105** (e.g., the host system **105-a**), or a host entity **210** thereof, with a particular public key or symmetric key. Such a mapping may also include a mapping between such keys and one or more partitions **245**, or a mapping between such keys or partitions **245** with one or more protection attributes, such as whether a partition **245** is configured with read protection, write protection, or both. In some examples, a mapping of the public key table **260** may include a mapping of a key, a host system **105**, or a host entity **210** with multiple partitions **245**, which may support each partition **245** using a common key but having a unique protection attribute. In some examples, the public key table may support a key hierarchy that allows a master host system **105**, or associated key, to assign partitions **245** to another host system **105** or to a host entity **210**, or their respective keys.

In some examples, the memory system **110-a** may include a platform configuration register (PCR) **270**, which may store or measure a software state (e.g., version, update status), such as a state of software running on the memory system **110-a**, and configuration data used by such software (e.g., to represent the platform software state of the memory system **110-a**). In some examples, the PCR **270** may include information that can be evaluated to determine whether the memory system **110-a** has been compromised or may be otherwise untrustworthy. Although the PCR **270** is illustrated separately from the memory system controller **115-a**, in some examples, the PCR **270** may be part of or otherwise associated with the memory system controller **115-a**, such as a location associated with firmware for the memory system **110-a** or the memory system controller **115-a** (e.g., a local memory **120**). Such techniques may support the PCR **270** storing or measuring a state of such firmware, which may be used to evaluate whether such firmware has been adversely updated (e.g., to evaluate whether the memory system **110-a** can be authenticated).

In some examples, the memory system **110-a** may include an RPMB **265**, which may be provided as a means to store data in an authenticated and replay protected manner, which may only be read and written via successfully authenticated read and write accesses. In some examples, the RPMB **265** may include information that can be evaluated to determine whether signaling exchanged with the memory system **110-a** has been intercepted and replayed, which may indicate whether one or more devices or connections of the system **200** are untrustworthy. Although the RPMB **265** is illustrated separately from the memory system controller **115-a**, in some examples, the RPMB **265** may be part of or otherwise associated with the memory system controller **115-a**, such as a storage location that includes firmware for the memory system **110-a** or the memory system controller **115-a** (e.g.,

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a local memory 120). In some examples, the RPMB 265 may be associated with a fixed size, a fixed set of addresses, or both.

In some examples, the memory system 110-a may include content 280, which may refer to various types of information stored at the memory system 110-a. In some examples, content 280 may be accessed or otherwise used to support various key generation (e.g., content-based key generation) or other cryptographic techniques in accordance with examples as disclosed herein. For example, the content 280 may include firmware of the memory system 110-a, such as boot code (e.g., first-stage boot code, “L0” boot code, second-stage boot code, “L1” boot code), or an FSD, which may establish an operating or cryptographic state of the memory system 110-a. In some examples, information associated with the content 280 may be used by the memory system 110-a to support various authentication or encryption techniques (e.g., to generate a certificate for operation with the host system 105-a). Although the content 280 is illustrated separately from the memory system controller 115-a, in some examples, the content 280 may be part of or otherwise associated with the memory system controller 115-a, such as a storage location that includes firmware for the memory system 110-a or the memory system controller 115-a. Further, although the content 280 is illustrated separately from the storage 240, the content 280 may, in some examples, refer to information that is included in a portion of the storage 240 (e.g., in a separate or dedicated partition 245). In some implementations, the content 280 may receive information from or may refer to one or more aspects of the PCR 270.

One or more components of the system 200 may be configured to implement asymmetric key distribution to establish authenticated signaling, encrypted signaling, or both between the host system 105-a and the memory system 110-a (e.g., in accordance with authenticated system identity), which may include an implementation of cryptographic security functionality directly in the memory system 110-a (e.g., leveraging capabilities of the memory system controller 115-a to support various techniques for asymmetric cryptography).

In some examples, such techniques may involve passing fundamentally public device identification information between the host system 105-a and the memory system 110-a that supports private authentication of the respective system (e.g., device-specific or hardware-specific authentication without trying to maintain secrecy or avoid exposure of exchanged private or secret keying material corresponding to respective devices). In some examples, such asymmetric cryptography may be utilized to derive equivalent or otherwise symmetric keys on each side of signaling exchange (e.g., at each of the host system 105-a and the memory system 110-a) using a common secret that is not itself communicated between the host system 105-a and the memory system 110-a, which may leverage efficiencies of symmetric key techniques for authenticated or encrypted signaling relative to asymmetric key techniques. In some examples, such techniques may be implemented to establish a virtual authenticated channel 205 between the host system 105-a and the memory system 110-a, which may be used to transmit signaling (e.g., encrypted signaling, unencrypted signaling) and associated signatures (e.g., asymmetric signatures such as elliptic curve digital signature algorithm (ECDSA) signatures, symmetric signatures such as hashed message authentication code (HMAC) signatures) between the host system 105-a and the memory system 110-a.

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In some examples, the system 200 may be configured to support a signing and verifying (e.g., authentication) of signaling between the host system 105-a and the memory system 110-a (e.g., in accordance with signed command signaling, signed request signaling, signed data signaling, or signed response signaling), which may be implemented to authenticate the transmitting system of such signaling, or to ensure that the signaling has not been altered before being received by a receiving system, or both. In accordance with such techniques, a receiving system may be able to evaluate received signaling to determine whether transmitted signaling was transmitted by an unverified or unauthorized transmitting system, or whether the transmitted signaling was altered or otherwise compromised. In some examples, such techniques may support a one-to-many security arrangement, since multiple receiving systems may be able to implement a same public key (e.g., of an asymmetric key pair) of the transmitting system that is associated with a single private key (e.g., of the asymmetric key pair) of the transmitting system. A key that is included in a same key pair as another key may be referred to as a counterpart key for the other key (e.g., a private key and a public key within an asymmetric key pair may be counterparts for each other, and two symmetric keys within a symmetric key pair may be counterparts for each other).

In some examples for signing and verifying signaling between the host system 105-a and the memory system 110-a, a signature for a given instance of signaling (e.g., a message, a command, a request, a data packet, a response) may be derived by hashing or otherwise processing the instance of signaling with a function (e.g., a hash function, a cryptographic hash algorithm) that receives, as an input, the instance of signaling and a private key associated with the transmitting system. The output of such a function (e.g., a signature, a hash digest) may be recreated using the same function with the same instance of signaling and either the same private key associated with the transmitting system or an associated public key (e.g., of an asymmetric key pair) associated with the transmitting system. In an example, for an instance of signaling associated with a 1-megabyte program operation, a hashing function based on the 1-megabytes of data and a private key may be a 256-bit signature or hash digest.

To support verifying the authenticity of the transmitting system, the transmitting system may transmit the instance of signaling along with the corresponding signature, which may be received by a receiving system. The receiving system may have received or otherwise generated the associated public key of the transmitting system and, accordingly, may generate a trial signature based on the received instance of signaling and the associated public key of the transmitting system. If the trial signature matches the received signature, the receiving system may determine that the transmitting system was authentic (e.g., that the instance of signaling is a transmission from a trusted system) and may continue with processing or otherwise performing a responsive action to the received instance of signaling. In some implementations, signature generation may be configured such that, even when an instance of signaling is the same, a generated signature will be different. In such implementations, signature generation and verification operations may be further based on a random number, a nonce, or a monotonic counter that is understood to both the transmitting system and the receiving system.

In some examples, the system 200 may be configured to support an encryption and decryption of signaling between the host system 105-a and the memory system 110-a (e.g.,

in accordance with encrypted signatures, encrypted command signaling, encrypted request signaling, encrypted data signaling, or encrypted response), which may be implemented to secure the contents of such signaling from being intercepted and interpreted or otherwise processed (e.g., to maintain integrity of the signaling itself). In accordance with such techniques, a transmitting system may encrypt instances of signaling for transmission using a key (e.g., of a symmetric key pair) known to the transmitting system, and a receiving system may decrypt received instances of such signaling using a key known to the receiving system (e.g., of the same symmetric key pair), which may be the same as the symmetric key known to the transmitting system, or may be otherwise equivalent or operable for such decryption. In some examples, such techniques may support a one-to-one security arrangement, since a symmetric key pair may only be understood to a single transmitting system and a single receiving system (e.g., when a symmetric key pair is based on unique identifiers of each of the transmitting system and the receiving system). However, some cryptographic techniques may support arrangements other than a one-to-one security arrangement, such as when symmetric keys are based on unique identifiers of more than two systems.

Some implementations of the described techniques may utilize asymmetric cryptography where a public key associated with the host system **105-a** may be uploaded to one or more memory systems **110** (e.g., the memory system **110-a**) without exposing a private key of the host system **105-a**, which may prevent an adverse actor from stealing the key and impersonating the real key holder (e.g., impersonating the host system **105-a**). Such techniques may also allow a public key to be replaced, which may be different than other techniques such as those related to a RPMB or a replay-protection monotonic counter (RPMC). In some examples, such asymmetric cryptography techniques may facilitate the use of public key infrastructure (PKI) techniques, where keys may be verified through a standardized digital certificate chain.

In some implementations, the exchange of public keys may support the generation of symmetric keys at each of the host system **105-a** and the memory system **110-a** using such techniques as a Diffie-Hellman key exchange or elliptic-curve techniques, so that a symmetric secret can be shared between device and host without exposing the private keys of the respective systems. In some implementations, an asymmetric Diffie-Hellman key exchange can be performed between the host system **105-a** and the memory system **110-a** to generate symmetric keys that are then used to enable better performance at the host system **105-a** or the memory system **110-a** for authentication, encryption, or both. Further, ephemeral symmetric keys can be derived using a same algorithm shared by the host system **105-a** and the memory system **110-a** to make it more difficult for an adverse actor to extract or replicate such keys, based on various techniques for duration-initiated or event-initiated generation of ephemeral keys.

In some examples, the exchange of public keys may be associated with the creation of digital certificates, which may include various signaling with or other interaction with one or more certificate authorities or registration authorities, or may involve self-signed certificates, or various combinations thereof. For example, the host system **105-a**, or a cloud authority or other centralized certification authority in communication with the host system **105-a**, may create a certificate signing request (CSR), which may be an example of a self-signed certificate that proves that memory system **110-a** has the private key associated with the public key in

the CSR. In some examples, such a CSR may be transferred from the memory system **110-a** to a centralized certification authority as part of a manufacturing operation (e.g., for manufacturing the memory system **110-a**). In some implementations, in response to an identity of the memory system **110-a** being confirmed (e.g., by a cloud authority), a manufacturer-endorsed certificate may be provided to the host system **105-a**, to the memory system **110-a**, or to both. In some examples, such techniques may support a requesting system downloading a manufacturer-endorsed certificate (e.g., a certificate endorsed by a certificate authority) or downloading the CSR.

FIG. 3 illustrates an example of a system **300** that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The system **300** may include one or more regions of a memory system, such as the region **305-a**, the region **305-b**, the region **305-c**, the region **305-d**, or the region **305-e**. In some cases, the memory system may be an example of aspects of the memory system **110** or the memory system **110-a**, as described with reference to FIGS. 1 and 2. In some cases, each of the one or more regions **305** may be a distinct region or group or memory cells, such as one or more blocks of memory cells, one or more memory dies, or other arrangements of memory cells. The memory cells included in a region **305** may be a continuous region of memory cells, or may be distributed throughout the memory system.

In some cases, a region **305** may be a protection region, such as a read-protected region. For example, the region **305-a** may be a read-protected region, and may be associated with or correspond to a key **310-a**. The key **310-a** may be a key associated with a host system or host entity, such as the host system **105-a** or a host entity **210** as described with reference to FIG. 2. In some cases, different host entities may use different keys. For example, a first host entity may be authorized to access the read-protected region **305-a**, and accordingly, the first host entity may use the key **310-a** when accessing the read-protected region **305-a**. Alternatively, a second host entity may be authorized to access the read-protected region **305-b**, and accordingly, the second host entity may use the key **310-b** when accessing the read-protected region **305-b**.

In some examples, multiple host entities may be authorized to access a same region **305**, such as the read-protected region **305-c**. In such examples, the read-protected region may be associated with multiple keys, such as a key **310-c** associated with a first host entity authorized to access the read-protected region **305-c** and a key **310-d** associated with a second host entity authorized to access the read-protected region **305-c**. In some cases, a single host entity may be associated with or use multiple keys **310**, and thus may be authorized to access multiple read-protected regions of the memory system.

In some cases, the host system and the memory system may use one or more symmetric key pairs. In such cases, both the host system and the memory system may store a shared key **310**. For example, is a host entity is authorized to access the read-protected region **305-a**, the host system may store the key **310-a** in the key storage **215**. Accordingly, the memory system may also store the key **310-a** in the key storage **250**. In some cases, the memory system may associate the key **310-a** with the host entity, for example in the public key table **260**, so that the memory system may verify commands sent from the host system.

Additionally or alternatively, the host system and the memory system may use asymmetric key pairs. In such cases, the host system may store a private key associated

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with a read-protected region, and the memory system may store a public key of the host system. For example, is a host entity is authorized to access the read-protected region 305-a, the host system may store the private key associated with the read-protected region 305-a in the key storage 215 and transmit the corresponding public key to the memory system. Accordingly, the memory system may store the key public key in the key storage 250. In some cases, key 310-a may be the public key received from the host system.

By way of example, a host entity may request data stored by the memory system in the read-protected region 305-a. Accordingly, the host system may generate a read command 315 to access the data. The host system may sign or encrypt the read command 315 using a key associated with the read-protected region 305-a (e.g., a shared key or a private key corresponding to a public key), such as a key associated with the host entity which requested the data. In some cases, signing the read command 315 may include performing a hash procedure to generating a hash (e.g., a hash digest) using the key associated with the read-protected region 305-a and the read command 315. In some cases, the signature of the host system may be the hash or result of the hashing procedure. In some cases, the host system may transmit the read command 315 and the signature to the memory system.

Additionally, or alternatively, the host system and the memory system may use asymmetric key pairs. In such cases, the host system may store a private key associated with a read-protected region, and the memory system may store a public key of the host system. For example, if a host entity is authorized to access the read-protected region 305-a, the host system may store the private key associated with the read-protected region 305-a in the key storage 215 and transmit the corresponding public key to the memory system. Accordingly, the memory system may store the key public key in the key storage 250. In some cases, key 310-a may be the public key received from the host system.

In some cases, the host system may request that the memory system signs the data associated with the read command 315. For example, the host system may transmit an additional command or indication along with the read command 315 to request the signed data. In such cases, after receiving the read command, the memory system may perform a hashing procedure to produce a hash using a key associated with the memory system (e.g., a private key of the memory system or a shared key between the host system and the memory system) and the retrieved data. The memory system may transmit the hash along with the data to the host system in response to the read command 315. The signature of the memory device may be the generated hash. The host system may verify or authenticate the data by performing a hashing procedure using a key associated with the host system (e.g., a private key of the host system or a shared key between the host system and the memory system) and the received data. If the result the hashing procedure matches the received signature, the host system may determine that the data was received from the memory system (e.g., the host system may determine that the data was not received from a separate or different memory system).

In some cases, the host entity which requested the data may not be authorized to access the read-protected region. For example, the host entity may be authorized to access data associated with the key 310-b, rather than the key 310-a. In such cases, the read command 315 may be signed using the key 310-b. Accordingly, after the memory system receives the read command 315 and signature, the memory system may determine that the has generated with the key

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310-a does not match the received signature. Accordingly, the memory system may return an indication that host system is not authorized to access the read-protected region 305-a. In some cases, the indication may include blank (e.g., all zeroes) or otherwise invalid (e.g., junk) data. In some examples (e.g., if the host system has requested that the memory system sign the data), the memory system may generate a signature for the requested data using a key associated with the memory system and transmit the signature to the host entity but not transmit the data to the host entity—thus, in some cases, a signature without any associated data may be an indication that host system is not authorized to access the read-protected region 305-a.

FIG. 4 illustrates an example of a process flow 400 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The process flow 400 may be implemented by a host system 405, for example using a host system controller, and by a memory system 410, for example using a controller, which may be examples of the respective devices described with reference to FIGS. 1 and 2. The memory system 410 may be an example of an eMMC system. In the following description of process flow 400, the operations may be performed in a different order than the order shown. For example, specific operations may also be left out of process flow 400, or other operations may be added to process flow 400.

In some examples, the process flow 400 may include a generation of public keys that may be communicated between the host system 405 and the memory system 410, which may be referred to as asymmetric keys or asymmetric public keys (e.g., public keys of an asymmetric key pair, public keys that each correspond to a respective private key of an asymmetric key pair). The generation of such public keys may be based on private keys maintained at the respective system, where such private keys are not shared outside the respective system. Such techniques may enable the host system 405 and the memory system 410 to sign various transmitted signaling (e.g., for authenticating a transmitting system), or to encrypt various transmitted signaling (e.g., for information integrity), or both without the exchange of private identification information unique to each system. Accordingly, such techniques may improve an ability to communicate with authenticity and integrity compared with other techniques where the distribution of cryptographic hardware identification information is not performed, or where such distribution may be more vulnerable to being cloned or stolen, such as techniques that distribute symmetric keys in a manner that may be insecure.

For example, the host system 405 may generate a host system public key, which may be based at least in part on (e.g., calculated using) a host system private key. In various examples, the host system private key may be stored at or otherwise generated at the host system 405 using a private identifier such as a fuse configuration, an identity stored in non-volatile memory, a PUF of the host system 405, or some other unique identifier of the host system 405, which may be protected from being cloned or extracted. In some examples, the host system public key may be associated with a particular host entity 210, or a particular range of addresses at the memory system 410 (e.g., a partition 245, or portion thereof), or a memory protection attribute (e.g., read protection, write protection), or a combination thereof.

In some examples, such public key attributes may not be associated with the host system public key generated, but may be later associated with one or more symmetric keys

that are generated based at least in part on the host system public key, or may be later assigned by the memory system 110-a.

In some examples, the process flow 400 may include an exchange of one or more keys between the host system 405 and the memory system 410. For example, at 415, the host system 405 may transmit one or more key associated with respective protected regions of the memory system 410, which may be received by the memory system 410. In some examples, (e.g., when the host system 405 and the memory system 410 are configured to support symmetric keys, encryption, or both), at 420, the memory system 410 may transmit one or more public keys for the memory system 410, which may be received by the host system 405. In some examples, the transmitted public keys may be stored at the respective receiving system (e.g., in key storage 215, in key storage 250), such as being stored in a non-volatile storage of or otherwise in communication with the respective receiving system. In some other examples, such transmitted or received public keys may not be stored, but keys generated based on such transmitted or received public keys may be stored upon further processing. In some examples, such asymmetric public keys may be updatable, where such updating may be initiated based on a timer or event, and such updated or superseding asymmetric public keys that are subsequently generated may accordingly be transmitted from a generating system to a receiving system.

In some examples, the process flow 400 may include a generation of symmetric keys by the host system 405 and the memory system 410, which may be calculated based at least in part on the respectively received public keys. Such symmetric keys may be generated to be equal or otherwise equivalent between the two systems (e.g., as a shared secret), or otherwise operable to for one to be used to authenticate information that has been signed using the other, or for one to be used to decode information that has been encoded using the other, or both, despite avoiding the transmission of private information. For example, at 425, the host system 405 may generate a symmetric key, which may be based at least in part on (e.g., calculated using) the one or more keys transmitted by the memory system 410 at 420, and the one or more keys of the host system 405. Further, at 430, the memory system 410 also may generate a symmetric key, which may be based at least in part on (e.g., calculated using) the one or more keys transmitted by the host system 405 at 415 and the one or more keys of the memory system 410. In some examples, such symmetric keys may be generated using Diffie-Hellmann techniques or other exponential key exchange or generation protocol, including elliptic-curve techniques. In some examples, the generated symmetric keys may be stored at the generating system (e.g., in key storage 215, in key storage 250), such as being stored in a non-volatile storage of or otherwise in communication with the respective generating system.

The process flow 400 may describe transmitting a read command from the host system 405 to the memory system 410 to access data stored at one or more protected regions of the memory system 410. For example, at 435, a command may be signed. For example, the host system 405 may generate a signed command to access data stored at a read-protected region of the memory system 410 (e.g., the read-protected region 305 as described with reference to FIG. 3), such as a read command. In some cases, generating the signed command may include generating a signature by performing a hash procedure using a key associated with the read-protected region of the memory system 410 and the read command and including the signature in the command.

Accordingly, the signed command may include an access command (e.g., the command to read data stored at the memory system 410) and the signature generated using the hash procedure.

At 440, the signed command may be transmitted to the memory system 410. For example, the host system 405 may transmit the signed command to the memory system 410. In some examples, the host system 405 may additionally transmit an indication that the signed command has been signed to the memory system 410. For example, the host system 405 may adjust a state of a channel between the host system 405 and the memory system 410 to indicate that the signed command is signed.

At 445, the signed command may be authenticated. For example, the memory system 410 may authenticate the signed command using the included signature. In some cases, authenticating the signed command may include performing a hash procedure or otherwise translating the signature included in the signed command using the key associated with the read-protected region of the memory system 410 to determine the identity of the host system 405. In some cases, the identity of the host system 405 may include an indication of a host entity which initiated the read command, such as a host entity 210 described with reference to FIG. 2.

In some cases, the key associated with the read-protected region of the memory system may be a key stored by the host system 405 (e.g., in the key storage 250, as described with reference to FIG. 2), such as a private key for the host system 405, which may be paired with a public key of the host system 405 (e.g., the private key and the public key may be part of an asymmetric key pair). The host system may transmit the public key to the memory system 410, for example over a virtual authenticated channel 205 as described with reference to FIG. 2. In such cases, the host system 405 may encrypt the read command at 475 using the private key for the host system 405. Accordingly, the memory system 410 may authenticate the read command at 445 using a public key for the host system 405.

Additionally or alternatively, the key associated with the read protected region of the shared key stored by the host system 405 (e.g., in the key storage 250, as described with reference to FIG. 2), which may be shared with a key of the memory system 410 (e.g., the key of the host system 405 and the key of the memory system 410 may be part of a symmetric key pair). The host system may transmit the shared key to the memory system 410, for example over a virtual authenticated channel 205 as described with reference to FIG. 2. In such cases, the host system 405 may encrypt the read command at 440 using the shared key. Accordingly, the memory system 410 may authenticate the read command at 445 using a shared key.

At 450, it may be determined whether the host system 405 is authorized to access the read-protected region of the memory system 410. For example, the memory system 410 may determine whether the signature of the signed command matches a signature generated by the hash procedure performed at 455. If the received signature and the generated signature match, the memory system 410 may determine that the host system 405 is authorized to access the data stored at the read-protected region. In such cases, the memory system 410 may retrieve the data requested in the read command from the read-protected region and may, at 455, transmit a response which includes the data to the host system 405.

Additionally or alternatively, the memory system at 450 may determine that the signature of the received signed

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command does not match the signature generated using the hash procedure performed at 445. For example, the read command may have been signed using a key associated with a different read-protected region (e.g., a second read-protected region). In such cases, the memory system 410 may, at 460, transmit a response to the host system 405 indicating that the host system 405 may not be authorized to access the data.

In some cases, the signed command may include a command for the memory system 410 to sign the data associated with signed command. In such cases, at 455, the response may be signed. For example, the memory system 410 may generate a signature using the data and a key associated with the memory system 410, such as private key for the memory system 410, or a shared key between the host system 405 and the memory system 410. In some cases, the signature may be included in the response and transmitted to the host system at 470. In some cases, if the memory system 410 has determined that the host system 405 may not be authorized to access the data, the response may include a signature generated using the key associated with the read-protected region.

Accordingly, at 465, the response may be authenticated. For example, the host system 405 may decrypt or otherwise translate the signature included in the response using a key for the host system 405, such as a private key for the host system 405, a shared key between the host system 405 and the memory system 410, or the key associated with the read-protected region. By decrypting the response, the host system 405 may determine the source of the data (e.g., the host system 405 may determine whether the data came from the memory system 410).

In some cases, the host system 405 may update the key associated with the read-protected region. For example, the host system 405 may generate a second key associated with the read-protected region of the memory system 410 and transmit the second key to the memory system 410. Accordingly, the memory system 410 may update the associated with the read-protected region, for example by updating the key storage 250, the public key table 260, or both, as described with reference to FIG. 2. In some cases, the host system 405 may update the key associated with the read protected region after an elapsed time period (e.g., a cryptoperiod).

In some cases, one or more read-protected regions of the memory system 410 may be updated. For example, the memory system 410 may adjust the size of one or more read-protected regions, such as by including more or fewer memory cells in an adjusted read protected region. Additionally or alternatively, the memory system 410 may adjust an address range of one or more read-protected regions. For example, as part of adjusting a read-protected region from a first address range to a second address range, the memory system 410 may transfer data (e.g., read-protected data) stored at the first address range to the second address range. In some cases, an updated read-protected region may be associated with a same key before and after being updated. In some cases, the host system 405 may transmit a command to the memory system 410 to update one or more read-protected regions.

Aspects of the process flow 400 may be implemented by a controller, among other components. Additionally or alternatively, aspects of the process flow 400 may be implemented as instructions stored in memory (e.g., firmware stored in a memory coupled with the memory system 410, the host system 405, or both). For example, the instructions,

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when executed by a controller, may cause the controller to perform the operations of the process flow 400.

FIG. 5 shows a block diagram 500 of a host device 520 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The host device 520 may be an example of aspects of a host device as described with reference to FIGS. 1 through 4. The host device 520, or various components thereof, may be an example of means for performing various aspects of authenticated reading of memory system data as described herein. For example, the host device 520 may include a key transmission component 525, a signaling transmission component 530, a response reception component 535, a key reception component 540, a key management component 545, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

The key transmission component 525 may be configured as or otherwise support a means for transmitting, by a host system to a memory system, one or more keys associated with the host system, where each of the one or more transmitted keys is for association with a respective protection region of one or more protection regions at the memory system. The signaling transmission component 530 may be configured as or otherwise support a means for transmitting, by the host system to the memory system, a signed command to read data from a first protection region of the one or more protection regions, where the signed command is signed based at least in part on a counterpart key corresponding to a first transmitted key of the one or more transmitted keys, and where the first protection region is associated with the first transmitted key associated with the host system. The response reception component 535 may be configured as or otherwise support a means for receiving the data from the memory system based at least in part on transmitting the signed command.

In some examples, the key reception component 540 may be configured as or otherwise support a means for receiving, at the host system, one or more public keys associated with the memory system, where the counterpart key includes a private key associated with the host system, each of the one or more received public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the one or more transmitted keys associated with the host system, and the signed command is signed further based at least in part on a first received public key of the one or more received public keys associated with the memory system.

In some examples, the key management component 545 may be configured as or otherwise support a means for generating a symmetric key based at least in part on the private key associated with the host system and the first received public key associated with the memory system, where the signed command is signed based at least in part on the generated symmetric key.

In some examples, the signaling transmission component 530 may be configured as or otherwise support a means for transmitting, from the host system to the memory system, an indication that the signed command is signed.

In some examples, the response reception component 535 may be configured as or otherwise support a means for receiving, from the memory system, the signature of the memory system in association with the data. In some examples, the response reception component 535 may be configured as or otherwise support a means for determining

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whether the data was received from the memory system based at least in part on the received signature of the memory system.

In some examples, the signature of the memory system is based at least in part on a key associated with the memory system.

In some examples, the key associated with the memory system includes a private key associated with the memory system. In some examples, determining whether the data was received from the memory system is further based at least in part on a public key corresponding to the private key associated with the memory system.

In some examples, the signaling transmission component 530 may be configured as or otherwise support a means for transmitting, by the host system to the memory system, a second signed command to read second data from a second protection region of the one or more protection regions at the memory system, where the second signed command is signed based at least in part on the counterpart key corresponding to the first transmitted key associated with the host system. In some examples, the response reception component 535 may be configured as or otherwise support a means for receiving, from the memory system, an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

In some examples, to support receiving the indication that the host system is not authorized to access the second protection region, the response reception component 535 may be configured as or otherwise support a means for receiving a signature based at least in part on the second data and a second key associated with the second protection region without receiving the second data.

In some examples, the signaling transmission component 530 may be configured as or otherwise support a means for transmitting, by the host system to the memory system, an unsigned command to read second data from the first protection region at the memory system. In some examples, the response reception component 535 may be configured as or otherwise support a means for receiving, from the memory system, an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

In some examples, the key management component 545 may be configured as or otherwise support a means for generating, after transmitting the signed command, an updated key associated with the host system and for association with the first protection region at the memory system. In some examples, the key transmission component 525 may be configured as or otherwise support a means for transmitting the updated key to the memory system. In some examples, the signaling transmission component 530 may be configured as or otherwise support a means for transmitting, by the host system to the memory system, a second signed command to read second data from the first protection region, where the second signed command is based at least in part on a second counterpart key corresponding to the updated key. In some examples, the response reception component 535 may be configured as or otherwise support a means for receiving the second data from the memory system based at least in part on transmitting the second signed command.

In some examples, transmitting the updated key is based at least in part on an elapsed time since transmitting the first transmitted key satisfying a threshold.

In some examples, the signaling transmission component 530 may be configured as or otherwise support a means for

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transmitting, to the memory system, an indication of a size of the first protection region, an address range corresponding to the first protection region, or any combination thereof.

FIG. 6 shows a block diagram 600 of a memory device 620 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The memory device 620 may be an example of aspects of a memory device as described with reference to FIGS. 1 through 4. The memory device 620, or various components thereof, may be an example of means for performing various aspects of authenticated reading of memory system data as described herein. For example, the memory device 620 may include a key reception component 625, a signaling reception component 630, a key management component 635, a response transmission component 640, a key transmission component 645, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

The key reception component 625 may be configured as or otherwise support a means for receiving, at a memory system from a host system, one or more keys associated with the host system, where each of the one or more received keys corresponds to a respective counterpart key associated with the host system and is for association with a respective protection region of one or more protection regions at the memory system. The signaling reception component 630 may be configured as or otherwise support a means for receiving, at the memory system, a signed command to read data from a first protection region of the one or more protection regions, where the first protection region is associated with a first received key of the one or more received keys. The key management component 635 may be configured as or otherwise support a means for determining whether to read the data from the first protection region based at least in part on attempting to decrypt a signature of the signed command, where attempting to decrypt the signature is based at least in part on the first received key associated with the host system. The response transmission component 640 may be configured as or otherwise support a means for transmitting the data to the host system based on least in part on successfully decrypting the signature of the signed command.

In some examples, the key transmission component 645 may be configured as or otherwise support a means for transmitting, to the host system, one or more public keys associated with the memory system, where each of the one or more transmitted public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the one or more received keys associated with the host system, and attempting to decrypt the signature of the signed command is further based at least in part on a first private key associated with the memory system.

In some examples, the key management component 635 may be configured as or otherwise support a means for generating a symmetric key based at least in part on the first received key associated with the host system and the first private key associated with the memory system, where attempting to decrypt the signature of the signed command is based at least in part on the generated symmetric key.

In some examples, the signaling reception component 630 may be configured as or otherwise support a means for receiving an indication that the signed command is signed.

In some examples, the response transmission component 640 may be configured as or otherwise support a means for transmitting a signature of the memory system in association

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with the data, the signature based at least in part on a key associated with the memory system.

In some examples, the signaling reception component **630** may be configured as or otherwise support a means for receiving, at the memory system, a second signed command to read second data from a second protection region of the one or more protection regions at the memory system, where the second signed command is signed based at least in part on a counterpart key corresponding to the first received key associated with the host system. In some examples, the key management component **635** may be configured as or otherwise support a means for determining whether to read the data from the second protection region based at least in part on attempting to decrypt the signature of the second signed command based at least in part on the first received key associated with the host system. In some examples, the response transmission component **640** may be configured as or otherwise support a means for transmitting, to the host system, an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

In some examples, the signaling reception component **630** may be configured as or otherwise support a means for receiving, at the memory system, an unsigned command to read second data from the first protection region at the memory system. In some examples, the response transmission component **640** may be configured as or otherwise support a means for transmitting, to the host system, an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

In some examples, the key management component **635** may be configured as or otherwise support a means for storing, at the memory system, each of the one or more received keys associated with the host system, an indication of the respective protection region associated with each of the one or more received keys associated with the host system, or any combination thereof.

In some examples, the signaling reception component **630** may be configured as or otherwise support a means for receiving, from the host system, an indication of a size of the first protection region, an address range corresponding to the first protection region, or any combination thereof.

In some examples, the memory system includes an eMMC device, a UFS device, a SD device, an SSD, or any combination thereof.

FIG. 7 shows a flowchart illustrating a method **700** that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The operations of method **700** may be implemented by a host device or its components as described herein. For example, the operations of method **700** may be performed by a host device as described with reference to FIGS. 1 through 5. In some examples, a host device may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally or alternatively, the host device may perform aspects of the described functions using special-purpose hardware.

At **705**, the method may include transmitting, by a host system to a memory system, one or more keys associated with the host system, where each of the one or more transmitted keys is for association with a respective protection region of one or more protection regions at the memory system. The operations of **705** may be performed in accordance with examples as disclosed herein. In some examples,

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aspects of the operations of **705** may be performed by a key transmission component **525** as described with reference to FIG. 5.

At **710**, the method may include transmitting, by the host system to the memory system, a signed command to read data from a first protection region of the one or more protection regions, where the signed command is signed based at least in part on a counterpart key corresponding to a first transmitted key of the one or more transmitted keys, and where the first protection region is associated with the first transmitted key associated with the host system. The operations of **710** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **710** may be performed by a signaling transmission component **530** as described with reference to FIG. 5.

At **715**, the method may include receiving the data from the memory system based at least in part on transmitting the signed command. The operations of **715** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **715** may be performed by a response reception component **535** as described with reference to FIG. 5.

In some examples, an apparatus as described herein may perform a method or methods, such as the method **700**. The apparatus may include, features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

Aspect 1: The apparatus, including features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., by a host system to a memory system) one or more keys associated with the host system, where each of the one or more transmitted keys is for association with a respective protection region of one or more protection regions at the memory system; transmitting (e.g., by the host system to the memory system) a signed command to read data from a first protection region of the one or more protection regions, where the signed command is signed based at least in part on a counterpart key corresponding to a first transmitted key of the one or more transmitted keys, and where the first protection region is associated with the first transmitted key associated with the host system; and receiving the data (e.g., from the memory system) based at least in part on transmitting the signed command.

Aspect 2: The apparatus of aspect 1, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving, at the host system, one or more public keys associated with the memory system, where the counterpart key includes a private key associated with the host system, each of the one or more received public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the one or more transmitted keys associated with the host system, and the signed command is signed further based at least in part on a first received public key of the one or more received public keys associated with the memory system.

Aspect 3: The apparatus of aspect 2, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for generating a symmetric key based at least in part on the private key associated with the host system and the first received public key associated with the memory system, where the signed command is signed based at least in part on the generated symmetric key.

Aspect 4: The apparatus of any of aspects 1 through 3, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., from the host system to the memory system) an indication that the signed command is signed.

Aspect 5: The apparatus of any of aspects 1 through 4, where the signed command indicates for the memory system to transmit a signature of the memory system in association with the data, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving (e.g., from the memory system) the signature of the memory system in association with the data and determining whether the data was received from the memory system based at least in part on the received signature of the memory system.

Aspect 6: The apparatus of aspect 5, where the signature of the memory system is based at least in part on a key associated with the memory system.

Aspect 7: The apparatus of aspect 6, where the key associated with the memory system is a private key associated with the memory system and determining whether the data was received from the memory system is further based at least in part on a public key corresponding to the private key associated with the memory system.

Aspect 8: The apparatus of any of aspects 1 through 7, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., by the host system to the memory system) a second signed command to read second data from a second protection region of the one or more protection regions at the memory system, where the second signed command is signed based at least in part on the counterpart key corresponding to the first transmitted key associated with the host system, and receiving (e.g., from the memory system) an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

Aspect 9: The apparatus of aspect 8 where operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving the indication that the host system is not authorized to access the second protection region, includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving a signature based at least in part on the second data and a second key associated with the second protection region without receiving the second data.

Aspect 10: The apparatus of any of aspects 1 through 9, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., by the host system to the memory system) an unsigned command to read second data from the first protection region at the memory system and receiving (e.g., from the memory system) an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

Aspect 11: The apparatus of any of aspects 1 through 10, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for generating, after transmitting the signed command, an updated key associated with the host system and for association with the first protection region at the memory system; transmitting the updated key (e.g., to the memory system); transmitting (e.g., by the host system to the memory system), a second signed command to read second data from the first protection region, where the second signed command is based at least in part a second counterpart key corresponding to the updated key; and receiving the

second data (e.g., from the memory system) based at least in part on transmitting the second signed command.

Aspect 12: The apparatus of aspect 11, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting the updated key is based at least in part on an elapsed time since transmitting the first transmitted key satisfying a threshold.

Aspect 13: The apparatus of any of aspects 1 through 12, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., to the memory system) an indication of a size of the first protection region, an address range corresponding to the first protection region, or any combination thereof.

FIG. 8 shows a flowchart illustrating a method 800 that supports authenticated reading of memory system data in accordance with examples as disclosed herein. The operations of method 800 may be implemented by a memory device or its components as described herein. For example, the operations of method 800 may be performed by a memory device as described with reference to FIGS. 1 through 4 and 6. In some examples, a memory device may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally or alternatively, the memory device may perform aspects of the described functions using special-purpose hardware.

At 805, the method may include receiving, at a memory system from a host system, one or more keys associated with the host system, where each of the one or more received keys corresponds to a respective counterpart key associated with the host system and is for association with a respective protection region of one or more protection regions at the memory system. The operations of 805 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 805 may be performed by a key reception component 625 as described with reference to FIG. 6.

At 810, the method may include receiving, at the memory system, a signed command to read data from a first protection region of the one or more protection regions, where the first protection region is associated with a first received key of the one or more received keys. The operations of 810 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 810 may be performed by a signaling reception component 630 as described with reference to FIG. 6.

At 815, the method may include determining whether to read the data from the first protection region based at least in part on attempting to decrypt a signature of the signed command, where attempting to decrypt the signature is based at least in part on the first received key associated with the host system. The operations of 815 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 815 may be performed by a key management component 635 as described with reference to FIG. 6.

At 820, the method may include transmitting the data to the host system based on least in part on successfully decrypting the signature of the signed command. The operations of 820 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 820 may be performed by a response transmission component 640 as described with reference to FIG. 6.

In some examples, an apparatus as described herein may perform a method or methods, such as the method 800. The

apparatus may include, features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

Aspect 14: The apparatus, including features, circuitry, logic, means, or instructions, or any combination thereof for receiving (e.g., at a memory system from a host system) one or more keys associated with the host system, where each of the one or more received keys corresponds to a respective counterpart key associated with the host system and is for association with a respective protection region of one or more protection regions at the memory system; receiving (e.g., at the memory system) a signed command to read data from a first protection region of the one or more protection regions, where the first protection region is associated with a first received key of the one or more received keys; determining whether to read the data from the first protection region based at least in part on attempting to decrypt a signature of the signed command, where attempting to decrypt the signature is based at least in part on the first received key associated with the host system; and transmitting the data (e.g., to the host system) based on least in part on successfully decrypting the signature of the signed command.

Aspect 15: The apparatus of aspect 14, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting (e.g., to the host system) one or more public keys associated with the memory system, where each of the one or more transmitted public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the one or more received keys associated with the host system, and attempting to decrypt the signature of the signed command is further based at least in part on a first private key associated with the memory system.

Aspect 16: The apparatus of aspect 15, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for generating a symmetric key based at least in part on the first received key associated with the host system and the first private key associated with the memory system, where attempting to decrypt the signature of the signed command is based at least in part on the generated symmetric key.

Aspect 17: The apparatus of any of aspects 14 through 16, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving an indication that the signed command is signed.

Aspect 18: The apparatus of any of aspects 14 through 17, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting a signature of the memory system in association with the data, the signature based at least in part on a key associated with the memory system.

Aspect 19: The apparatus of any of aspects 14 through 18, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving (e.g., at the memory system) a second signed command to read second data from a second protection region of the one or more protection regions at the memory system, where the second signed command is signed based at least in part on a counterpart key corresponding to the first received key associated with the host system; determining whether to read the data from the second protection region based at least in part on attempting to decrypt the signature of the second signed command based at least in part on the

first received key associated with the host system; and transmitting (e.g., to the host system) an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

Aspect 20: The apparatus of any of aspects 14 through 19, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving (e.g., at the memory system) an unsigned command to read second data from the first protection region at the memory system and transmitting (e.g., to the host system) an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

Aspect 21: The apparatus of any of aspects 14 through 20, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for storing, at the memory system, each of the one or more received keys associated with the host system, an indication of the respective protection region associated with each of the one or more received keys associated with the host system, or any combination thereof.

Aspect 22: The apparatus of any of aspects 14 through 21, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for receiving (e.g., from the host system) an indication of a size of the first protection region, an address range corresponding to the first protection region, or any combination thereof.

Aspect 23: The apparatus of any of aspects 14 through 22, where the memory system includes an eMMC device, a UFS device, a SD device, an SSD, or any combination thereof.

It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.

Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, the signal may represent a bus of signals, where the bus may have a variety of bit widths.

The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

The term “coupling” refers to a condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals are capable of being communicated between components over the conductive path. If a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other if the switch is open. If a controller isolates two components, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

The terms “if,” “when,” “based on,” or “based at least in part on” may be used interchangeably. In some examples, if the terms “if,” “when,” “based on,” or “based at least in part on” are used to describe a conditional action, a conditional process, or connection between portions of a process, the terms may be interchangeable.

The term “in response to” may refer to one condition or action occurring at least partially, if not fully, as a result of a previous condition or action. For example, a first condition or action may be performed and second condition or action may at least partially occur as a result of the previous condition or action occurring (whether directly after or after one or more other intermediate conditions or actions occurring after the first condition or action).

The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some examples, the substrate is a semiconductor wafer. In some other examples, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals. The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as an n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” if a voltage greater

than or equal to the transistor’s threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” if a voltage less than the transistor’s threshold voltage is applied to the transistor gate.

The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a hyphen and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

For example, the various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

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Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read-only memory (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A host system, comprising:

a host controller configured to couple with a memory system controller of a memory system, wherein the memory system is external to the host system, and wherein the host controller is configured to cause the host system to:

transmit, to the memory system, a plurality of public keys associated with the host system, wherein each of the plurality of transmitted public keys is for association with a respective protection region of a plurality of protection regions at the memory system; transmit a signed command to read data from a first protection region of the plurality of protection regions, wherein the signed command is signed based at least in part on a private counterpart key associated with the host system, wherein the private counterpart key and a first transmitted public key of the plurality of transmitted public keys comprise an asymmetric key pair, wherein the signed command comprises an access command and a signature generated using a function that receives, as inputs, the private counterpart key and the access command, wherein the access command indicates an access operation to be performed at the memory system, and wherein the first protection region is associated with the first transmitted public key associated with the host system;

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receive the data based at least in part on transmitting the signed command; and

transmit a command to adjust the first protection region from a first size to a second size, to adjust the first protection region from a first address range to a second address range, or both, wherein the adjusted first protection region is associated with the first transmitted public key associated with the host system.

2. The host system of claim 1, wherein the host controller is further configured to cause the host system to:

receive, at the host system, one or more public keys associated with the memory system, wherein:

each of the one or more received public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the plurality of transmitted public keys associated with the host system; and

the signed command is signed further based at least in part on a first received public key of the one or more received public keys associated with the memory system.

3. The host system of claim 2, wherein the host controller is further configured to cause the host system to:

generate a symmetric key based at least in part on the private counterpart key associated with the host system and the first received public key associated with the memory system, wherein the signed command is signed based at least in part on the generated symmetric key.

4. The host system of claim 1, wherein the host controller is further configured to cause the host system to:

transmit an indication that the signed command is signed.

5. The host system of claim 1, wherein the signed command indicates for the memory system to transmit a second signature of the memory system in association with the data, and wherein the host controller is further configured to cause the host system to:

receive the second signature of the memory system in association with the data; and

determine whether the data was received from the memory system based at least in part on the received second signature of the memory system.

6. The host system of claim 5, wherein the signature of the memory system is based at least in part on a key associated with the memory system.

7. The host system of claim 6, wherein:

the key associated with the memory system comprises a private key associated with the memory system, and determining whether the data was received from the memory system is further based at least in part on a public key corresponding to the private key associated with the memory system.

8. The host system of claim 1, wherein the host controller is further configured to cause the host system to:

transmit a second signed command to read second data from a second protection region of the plurality of protection regions at the memory system, wherein the second signed command is signed based at least in part on the private counterpart key corresponding to the first transmitted public key associated with the host system; and

receive an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

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9. The host system of claim 8, wherein, to receive the indication that the host system is not authorized to access the second protection region, the host controller is configured to cause the host system to:

receive a second signature of the memory system based at least in part on the second data and a second public key associated with the second protection region without receiving the second data.

10. The host system of claim 1, wherein the host controller is further configured to cause the host system to:

transmit an unsigned command to read second data from the first protection region at the memory system; and receive an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

11. The host system of claim 1, wherein the host controller is further configured to cause the host system to:

generate, after transmitting the signed command, an updated public key associated with the host system and for association with the first protection region at the memory system;

transmit the updated public key;

transmit a second signed command to read second data from the first protection region, wherein the second signed command is based at least in part on a second private counterpart key corresponding to the updated public key; and

receive the second data based at least in part on transmitting the second signed command.

12. The host system of claim 11, wherein the host controller is configured to cause the host system to:

transmit the updated public key based at least in part on an elapsed time since the first transmitted public key was transmitted satisfying a threshold.

13. A memory system, comprising:

a memory system controller configured to couple with a host system controller of a host system and cause the memory system to:

receive, from the host system, a plurality of public keys associated with the host system, wherein the host system is external to the memory system, wherein each of the plurality of received public keys corresponds to a respective private counterpart key associated with the host system and comprises an asymmetric key pair with the respective private counterpart key, and wherein each of the plurality of received public keys is for association with a respective protection region of a plurality of protection regions at the memory system;

receive a signed command to read data from a first protection region of the plurality of protection regions, wherein the signed command comprises an access command and a signature generated using a function that receives, as inputs, the respective private counterpart key and the access command, wherein the access command indicates an access operation to be performed at the memory system, and wherein the first protection region is associated with a first received public key of the plurality of received public keys;

determine whether to read the data from the first protection region based at least in part on attempting to decrypt the signature of the signed command, wherein attempting to decrypt the signature is based at least in part on the first received public key associated with the host system;

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transmit the data based on least in part on successfully decrypting the signature of the signed command; and receive a command to adjust the first protection region from a first size to a second size, to adjust the first protection region from a first address range to a second address range, or both, wherein the adjusted first protection region is associated with the first received public key associated with the host system.

14. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

transmit one or more public keys associated with the memory system, wherein:

each of the one or more transmitted public keys associated with the memory system corresponds to a respective private key associated with the memory system and is different than each of the plurality of received public keys associated with the host system; and

attempt to decrypt the signature of the signed command is further based at least in part on a first private key associated with the memory system.

15. The memory system of claim 14, wherein the memory system controller is further configured to cause the memory system to:

generate a symmetric key based at least in part on the first received public key associated with the host system and the first private key associated with the memory system, wherein attempting to decrypt the signature of the signed command is based at least in part on the generated symmetric key.

16. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

receive an indication that the signed command is signed.

17. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

transmit a second signature of the memory system in association with the data, the second signature based at least in part on a key associated with the memory system.

18. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

receive a second signed command to read second data from a second protection region of the plurality of protection regions at the memory system, wherein the second signed command is signed based at least in part on a private counterpart key corresponding to the first received public key associated with the host system; determine whether to read the data from the second protection region based at least in part on attempting to decrypt the signature of the second signed command based at least in part on the first received public key associated with the host system; and

transmit an indication that the host system is not authorized to access the second protection region based at least in part on the second signed command.

19. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

receive an unsigned command to read second data from the first protection region at the memory system; and transmit an indication that the host system is not authorized to access the first protection region based at least in part on the unsigned command.

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20. The memory system of claim 13, wherein the memory system controller is further configured to cause the memory system to:

store, at the memory system, each of the plurality of received public keys associated with the host system, an indication of the respective protection region associated with each of the plurality of received public keys associated with the host system, or any combination thereof.

21. The memory system of claim 13, wherein the memory system comprises an embedded multimedia card (eMMC) device, a universal flash storage (UFS) device, a secure digital (SD) device, a solid state drive (SSD), or any combination thereof.

22. A non-transitory computer-readable medium storing code comprising instructions which, when executed by a processor of a host system, the host system configured to be coupled with a memory system, cause the host system to:

transmit, to the memory system, a plurality of public keys associated with the host system, wherein each of the plurality of transmitted public keys is for association with a respective protection region of a plurality of protection regions at the memory system;

transmit a signed command to read data from a first protection region of the plurality of protection regions, wherein the signed command is signed based at least in part on a private counterpart key associated with the host system, wherein the private counterpart key and a first transmitted public key of the plurality of transmitted public keys comprise an asymmetric key pair, wherein the signed command comprises an access command and a signature generated using a function that receives, as inputs, the private counterpart key and the access command, wherein the access command indicates an access operation to be performed at the memory system, and wherein the first protection region is associated with the first transmitted public key associated with the host system;

receive the data based at least in part on transmitting the signed command; and

transmit a command to adjust the first protection region from a first size to a second size, to adjust the first protection region from a first address range to a second

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address range, or both, wherein the adjusted first protection region is associated with the first transmitted public key associated with the host system.

23. A non-transitory computer-readable medium storing code comprising instructions which, when executed by a processor of a memory system, the memory system configured to be coupled with a host system, cause the memory system to:

receive, from the host system, a plurality of public keys associated with the host system, wherein the host system is external to the memory system, wherein each of the plurality of received public keys corresponds to a respective private counterpart key associated with the host system and comprises an asymmetric key pair with the respective private counterpart key, and wherein each of the plurality of received public keys is for association with a respective protection region of a plurality of protection regions at the memory system;

receive a signed command to read data from a first protection region of the plurality of protection regions, wherein the signed command comprises an access command and a signature generated using a function that receives, as inputs, the respective private counterpart key and the access command, wherein the access command indicates an access operation to be performed at the memory system, and wherein the first protection region is associated with a first received public key of the plurality of received public keys;

determine whether to read the data from the first protection region based at least in part on attempting to decrypt the signature of the signed command, wherein attempting to decrypt the signature is based at least in part on the first received public key associated with the host system;

transmit the data based on least in part on successfully decrypting the signature of the signed command; and receive a command to adjust the first protection region from a first size to a second size, to adjust the first protection region from a first address range to a second address range, or both, wherein the adjusted first protection region is associated with the first received public key associated with the host system.

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